Introduction to Stochastic Integration

M. E. Caballero

March 22, 2017

Contents

Chapters **Page**

1 Introduction to conditional expectation **3**
1.1 Introduction and first results . . . 3
1.2 First properties of the conditional expectation . . . 7
1.3 Second group of properties. . . . 10
1.4 Relationship with independence . . . 11
1.5 Exercises . . . 14
1.6 Appendix . . . 16

2 Stochastic processes **17**
2.1 Exercises . . . 21

3 Martingales **23**
3.1 Some important properties of Martingales . . . 24
3.2 Exercises . . . 25

4 Brownian Motion **27**
4.1 Gaussian Random Variables . . . 27
4.1.1 **Some known results on gaussian vectors** . . . 27
4.1.2 **Gaussian processes** . . . 28
4.2 The Brownian Motion . . . 29
4.2.1 Computation of joint probabilities. . . . 32
4.2.2 Markov Property . . . 33
4.3 Exercises . . . 34

5 Stochastic Itô Integral **35**
5.1 Discrete case: Integration w.r.t. a Simple symmetric Random Walk . . . 35
5.2 Quadratic variation of Brownian motion. . . . 37
5.3 Stochastic Itô Integral . . . 38
5.4 Exercises. . . . 44
5.5 Itô Formulas . . . 45
5.6 Itô Process . . . 48
5.7 Integration by parts formula . . . 49
5.8 Exercises . . . 50

6 Stochastic Differential Equations **51**
6.1 Weak Solutions . . . 55
6.2 Exercises . . . 57

Bibliography **59**

Chapter 1

Introduction to conditional expectation

1.1 Introduction and first results

Conditional probability and conditional expectation are amongst of the most useful tools in probability theory and stochastic processes. We give here the main results that will be used in the course.
The difficult parts will be marked with * and can be skipped in the first reading of this text.

- ♢ A **probability space** $(\Omega, \mathcal{F}, \mathbb{P})$ consists of a nonempty set Ω, a **σ-algebra** of subsets of Ω, denoted by $\mathcal{F}$ and a probability measure $\mathbb{P}$ defined in $\mathcal{F}$.

- ♢ We denote by $\mathbb{R}$ the set of **real numbers** with the usual distance $d(x, y) = |x-y|,\ x, y \in \mathbb{R}$ and $\mathbb{R}^n$ the euclidean space

$$\{(x_1, x_2, \ldots, x_n) : x_i \in \mathbb{R}, i = 1, 2, \ldots, n\}$$

  with the usual norm.

- ♢ The **Borel σ-algebra** in any metric space M is the σ-algebra generated by the open sets of the space. We will denote it by $\mathcal{B}(M)$. The most frequently used in this text are $\mathcal{B}(\mathbb{R})$ and $\mathcal{B}(\mathbb{R}^n)$.

- ♢ A **random variable** X in $(\Omega, \mathcal{F})$ (or in $(\Omega, \mathcal{F}, \mathbb{P})$), is a function $X : \Omega \to \mathbb{R}$ which is $\mathcal{F}$-measurable, i.e. it verifies the condition $(X \in B) := X^{-1}(B) \in \mathcal{F}$ for all Borel subsets B of $\mathbb{R}$.

- ♢ The elements of a Borel σ-algebra are called **Borel sets**.

- ♢ If we have two metric spaces, M_1, M_2 and a function $F : M_1 \to M_2$ that is measurable with respect to the corresponding Borel σ-algebras, i.e. $(F \in B) := F^{-1}(B) \in \mathcal{B}(M_1)$, for all $B \in \mathcal{B}(M_2)$ is called a **Borel function**.

- ♢ $X : \Omega \to \mathbb{R}$ is called **integrable** in $(\Omega, \mathcal{F}, \mathbb{P})$ if $\int |X| d\,\mathbb{P} = \mathbb{E}(|X|) < \infty$.

- ♢ For $n > 1$, $n \in \mathbb{N}$, we will also use functions with values in the euclidean space $\mathbb{R}^n$. They are called **random vectors** in $(\Omega, \mathcal{F})$, if $X : \Omega \to \mathbb{R}^n$ is **$\mathcal{F}$-measurable**, i.e. it verifies

$$(X \in B) := X^{-1}(B) \in \mathcal{F}, \quad B \in \mathcal{B}(\mathbb{R}^n).$$

◇ $L_1(\Omega, \mathcal{F}, \mathbb{P})$ is the set (of equivalence clases with respect to the relation equal almost surely) of integrable random variables in $(\Omega, \mathcal{F}, \mathbb{P})$.

The idea behind the conditional expectation is that, in many applications and theoretical developements, we need to construct new random variables based on less information, i.e. measurable with respect to a smaller σ-algebra $\mathcal{G}$ and at the same time as close as possible to the original ones. This is properly defined as:

Definition 1.1. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{G}$ a sub-σ-algebra of $\mathcal{F}$.*

1. *Let X be an integrable random variable in $(\Omega, \mathcal{F}, \mathbb{P})$. The* **conditional expectation** *of X given $\mathcal{G}$ is a function $Z : \Omega \to \mathbb{R}$ such that*
   (a) *Z is $\mathcal{G}$-measurable*
   (b) *For all $D \in \mathcal{G}$, the equality $\int_D X d\mathbb{P} = \int_D Z d\mathbb{P}$ is verified.*
   (c) *The conditional expectation of X given $\mathcal{G}$ is denoted by $\mathbb{E}(X|\mathcal{G})$*
2. *If $A \in \mathcal{F}$, the* **conditional probability** *of A given $\mathcal{G}$ is a function $Z : \Omega \to R$ such that*
   (a) *Z is $\mathcal{G}$-measurable*
   (b) *For all $D \in \mathcal{G}$ the equality $\mathbb{P}(A \cap D) = \int_D Z d\mathbb{P}$ holds true.*
   (c) *The conditional probability of A given $\mathcal{G}$ is denoted by $\mathbb{P}(A|\mathcal{G})$.*

Remarks 1.2.

1. *We can also say in the former definition that Z is a random variable in $(\Omega, \mathcal{G})$ verifying the second condition.*
2. *Z is not unique, since any other $\mathcal{G}$-measurable function $Z' : \Omega \to \mathbb{R}$ and such that $Z = Z'$ almost surely verifies the second condition in both cases.*
3. *From the given definitions if $Z = \mathbb{P}(A|\mathcal{G})$ and $Z' = \mathbb{E}(\mathbb{1}_A|\mathcal{G})$, we have that $Z = Z'$ a.s. since for all $D \in \mathcal{G}$,*
$$\int_D Z d\mathbb{P} = \mathbb{P}(A \cap D) = \int_D \mathbb{1}_A d\mathbb{P} = \int_D Z' d\mathbb{P},$$
   and we know that two $\mathcal{G}$ measurable functions Z and Z' which integrate the same over all $D \in \mathcal{G}$, are equal a.s.
4. *Once these definitions are given, it is natural to look for the relationship between them and the ones given in the elementary probability courses. For instance, for two events A, B, the definition of the conditional probability of A given B when B has positive probability is simply:*
$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$
   and in the case when we have two random variables X, Y with joint density the conditional expectation of X given Y is defined directly with the use of this joint density.
5. *The relationship between these concepts is given in the exercises.*

First we want to find out if there exists random variables with the conditions mentioned in the definition and how can we construct them explicitly. One of the surprises we will have, is the difficulty to find the explicit form these conditional expectations or probabilities, except in the elementary cases. Another one is to find out that the proof of the existence requires of advanced theoretical results in measure theory. The result we will need is the Radon-Nikodym theorem for finite measures (Bartle Chapter 3, Page. 85).

Theorem 1.3 (Existence Theorem)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, X is a random variable in $L_1(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G}$ a sub-σ-algebra of $\mathcal{F}$. There exists a function $Z : \Omega \to \mathbb{R}$ such that*

1. *Z es $\mathcal{G}$-measurable,*
2. *Z is integrable,*
3. *for all $D \in \mathcal{G}$, we have $\int_D X d\mathbb{P} = \int_D Z d\mathbb{P}$,*
4. *If Z' is another function verifying the three former conditions, $Z = Z'$ $\mathbb{P}$-a.s.*

Proof. We first suppose that $X \geq 0$. We define $Q : \mathcal{G} \to \mathbb{R}$ as

$$Q(D) := \int_D X d\mathbb{P}.$$

Then Q is a finite measure on $\mathcal{G}$; to see this let $(A_n)_{n\in\mathbb{N}}$ be a pairwise disjoint sequence of sets in $\mathcal{G}$ and $A = \bigcup_{n\in\mathbb{N}} A_n$. We then have $\mathbb{1}_A = \sum_{n\in\mathbb{N}} \mathbb{1}_{A_n} = \lim_{N\to\infty} \sum_{n=1}^{N} \mathbb{1}_{A_n}$. The result is obtained by an application of the monotone convergence theorem. It is also easy a standard result in measure theory that if $\mathbb{P}(D) = 0$ then $Q(D) = 0$, so that Q is absolutely continuous with respect to P restricted to $\mathcal{G}$.
We can thus apply the Radon Nikodym to assure that there exists a positive random variable Z in $(\Omega, \mathcal{G})$ verifying

$$Q(D) = \int_D Z d\mathbb{P}, \;\; D \in \mathcal{G}.$$

It is clear that Z is the function we need since it satisfies all the stated conditions and it is unique a.s.

When X is integrable, we use the usual decomposition $X = X^+ - X^-$ and we apply the former result to each part of the decomposition to obtain Z^+ and Z^- with the stated properties and $Z = Z^+ - Z^-$ is the desired function.
The proof of the existence of the conditional probability follows the same idea or else we can do it by using Observations 1.2-(3). ■

Before the main subject is treated, we will recall several elementary but important results of measure theory.

Definition 1.4.

1. *If Y is a random variable in $(\Omega, \mathcal{F}, \mathbb{P})$, we define the* **law** *of Y as the probability measure defined in $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ by*

$$P_Y(B) = \mathbb{P}(Y \in B) = \mathbb{P}(Y^{-1}(B)), \;\; B \in \mathcal{B}(\mathbb{R})$$

2. **The σ-algebra generated** *by Y is*

$$\sigma(Y) := \{G \subseteq \Omega : \textit{such that there exists } B \in \mathcal{B}(\mathbb{R}) \textit{ with } G = Y^{-1}(B)\} = Y^{-1}(\mathcal{B}(\mathbb{R}))$$

It is left as an exercise to the reader, to verify that P_Y is a probability measure defined in the Borel sets of $\mathbb{R}$, that $\sigma(Y)$ is a σ-algebra in Ω, contained in $\mathcal{F}$ and that $\sigma(Y) \subset \mathcal{F}$ is the smallest σ-algebra with respect to which Y is measurable.

With this new measure we can prove the **change of variable formula**:

Proposition 1.5. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, Y a random variable in $(\Omega, \mathcal{F}, \mathbb{P})$ and $F\colon \mathbb{R} \to \mathbb{R}$ a Borel function. Then $F(Y) \in L_1(\Omega, \mathcal{F}, \mathbb{P})$ if and only if $F \in L_1(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_Y)$, and*

1. $\mathbb{E}\left(F(Y)\right) = \int F(Y) d\mathbb{P} = \int_{\mathbb{R}} F(y) dP_Y.$
2. *When P_Y has density $q\colon \mathbb{R} \to \mathbb{R}^+$ then*

$$\mathbb{E}(F(Y)) = \int F(Y) d\mathbb{P} = \int_{\mathbb{R}} F(y) dP_Y = \int_{\mathbb{R}} F(y) q(y) dy$$

Proof. For $B \in \mathcal{B}(\mathbb{R})$ and $F = \mathbb{1}_B$, the result is immediate, since

$$F(Y) = \mathbb{1}_{(Y \in B)}.$$

Then the result follows (by using linearity on both sides) for simple measurable functions and by the usual procedure (using the monotone convergence theorem) we obtain it for positive measurable functions. Now for any $F \in L_1(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_y)$ using the decomposition $F = F^+ - F^-$ and applying the former result to each term ends the proof. ■

Proposition 1.6. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X, Y two random variables in $(\Omega, \mathcal{F})$ and $\sigma(Y)$ the σ-algebra generated by Y. Then X is $\sigma(Y)$-measurable if and only if there exists a Borel function $F\colon \mathbb{R} \to \mathbb{R}$ such that $X = F(Y)$. The same result holds true for random vectors in $\mathbb{R}^n$ and $F\colon \mathbb{R}^n \to \mathbb{R}^n$.*

Proof. It is immediate that if $X = F(Y)$ then for any Borel subset B of $\mathbb{R}$, $(F^{-1}(B)) \in \mathcal{B}(\mathbb{R})$ and $X^{-1}(B) = Y^{-1}(F^{-1}(B)) \in \sigma(Y)$. This proves that X is $\sigma(Y)$-measurable. To see the other implication, we will first suppose that X is a simple $\sigma(Y)$-measurable function. Let $a_1, a_2, \dots a_k$ be the different values of X and $A_i = X^{-1}(\{a_i\}) = \{X = a_i\}$; it is clear that $(A_i)_{i=1}^k$ is a $\sigma(Y)$- measurable partition of Ω. This implies, by definition of $\sigma(Y)$, that for each $i = 1, 2, \dots, k$, there exists a Borel set B_i of $\mathbb{R}$ such that $A_i = Y^{-1}(B_i) = \{Y \in B_i\}$

$$X(w) = \sum_{i=1}^k a_i \mathbb{1}_{A_i}(\omega) = \sum_{i=1}^k a_i \mathbb{1}_{\{Y \in B_i\}}(\omega) = \sum_{i=1}^k a_i \, \mathbb{1}_{B_i}(Y(\omega))$$

The Borel function $F\colon \mathbb{R} \to \mathbb{R}$ we need is:

$$F(x) = \sum_{i=1}^k a_i \mathbb{1}_{B_i}(x).$$

Now let X be a $\sigma(Y)$- measurable and positive function. It is well known that there exists a non decreasing sequence $(X_n)_{n \in \mathbb{N}}$ of simple $\sigma(Y)$-measurable functions such that for $\omega \in \Omega$, $X(\omega) = \lim_{n \to \infty} X_n(\omega)$. For each $n \in \mathbb{N}$ let F_n the Borel function which exists (by the first part of this proof) and verifies $X_n = F_n(Y)$. Now define $F(x), \ x \in \mathbb{R}$ as

$$\limsup_{n \to \infty} F_n$$

It is clear that F is also a Borel function and $X = \lim_{n\to\infty} X_n = \lim_{n\to\infty} F_n(Y) = (\limsup_{n\to\infty} F_n)(Y) = F(Y)$. Now for an arbitrary random variable X, we use the decomposition $X = X^+ - X^-$, and apply the first part of the proof to each term. ■

Observation 1.7. *This property will prove very important for the case when the σ-algebra $\mathcal{G}$ used to condition is $\sigma(Y)$ for some random variable Y (see definition 1.4) In this situation we have, thanks to the former result, that there will exist a Borel function $g\colon \mathbb{R} \to \mathbb{R}$ such that*

$$E(X|Y) := E(X|\sigma(Y)) = g(Y).$$

Definition 1.8. *The Borel function $g\colon \mathbb{R} \to \mathbb{R}$ such that $g(Y) = E(X|\sigma(Y))$ is known as* **the evaluated conditional expectation** *of X given Y and it is denoted*

$$g(y) = \mathbb{E}(X|Y = y) \quad y \in \mathbb{R}$$

and g is unique P_Y-a.s.

We will study three groups of properties of the conditional expectation: the first the ones similar to the properties of the expectation, such a linearity, monotonicity, convergence theorems and some inequalities. The second group will include those related to the dependence of the σ-algebra(s) used to condition, and finally the relationship with independence.

1.2 First properties of the conditional expectation

Theorem 1.9. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G}$ be a sub-σ-algebra of $\mathcal{F}$ and for $n = 1, 2, \ldots,$ $X, X_n\colon (\Omega, \mathcal{F}, \mathbb{P}) \to \mathbb{R}$, are random variables*

1. *If $c \in \mathbb{R}$ and $X = c$ then $\mathbb{E}(X|\mathcal{G}) = c$.*
2. *If $X \geq 0$ a.s. then $\mathbb{E}(X|\mathcal{G}) \geq 0$ a.s.*
3. *If $a, b \in \mathbb{R}$, $\mathbb{E}(aX_1 + bX_2|\mathcal{G}) = a\,\mathbb{E}(X_1|\mathcal{G}) + b\,\mathbb{E}(X_2|\mathcal{G})$.*
4. *(Monotony) If $X_1 \geq X_2$ then $E(X_1|\mathcal{G}) \geq E(X_2|\mathcal{G})$.*
5. *(Conditional monotone convergence theorem: CMCT).*
   If $0 \leq X_n \leq X_{n+1}$, $n = 1, 2, \ldots$ and $X = \lim_{n\to\infty} X_n$ then
   $$\mathbb{E}(X|\mathcal{G}) = \lim_{n\to\infty} \mathbb{E}(X_n|\mathcal{G}).$$
6. *(Conditional dominated convergence theorem: CDCT).*
   If $X = \lim_{n\to\infty} X_n$ and there exists a $Y \in L_1(\Omega, \mathcal{F}, \mathbb{P})$ such that $|X_n| \leq Y$ a.s. for all $n = 1, 2, \ldots$ then $X \in L_1(\Omega, \mathcal{F}, \mathbb{P})$ and
   $$\mathbb{E}(X|\mathcal{G}) = \lim_{n\to\infty} \mathbb{E}(X_n|\mathcal{G})$$
7. *(Conditional Fatou lemma: CFL).*
   If $0 \leq X_n$, $n = 1, 2, \ldots$
   $$\mathbb{E}(\liminf_n X_n|\mathcal{G}) \leq \liminf_n \mathbb{E}(X_n|\mathcal{G})$$

8. *(Conditional Jensen inequality).*
   If $X \in L_1(\Omega, \mathcal{F}, \mathbb{P})$ and $g\colon \mathbb{R} \to \mathbb{R}$ is convex then
$$g(\mathbb{E}(X|\mathcal{G})) \leq \mathbb{E}(g(X)|\mathcal{G}).$$

9. *Some particular cases of Jensen's inequality are:*
   (a) $|\mathbb{E}(X|\mathcal{G})| \leq \mathbb{E}(|X||\mathcal{G})$
   (b) $(\mathbb{E}(X|\mathcal{G}))^+ \leq \mathbb{E}(X^+|\mathcal{G})$ *and* $(\mathbb{E}(X|\mathcal{G}))^- \leq \mathbb{E}(X^-|\mathcal{G})$
   (c) $|\mathbb{E}(X|\mathcal{G})|^p \leq \mathbb{E}(|X|^p|\mathcal{G})$, $p \geq 1$

All equalities or inequalities must be considered as $\mathbb{P}$-a.s.

Proof.

1. It is immediate because the constant function is measurable with respect to any σ-algebra.

2. Since $X \geq 0$ we have
$$0 \leq \int_D X d\,\mathbb{P} = \int_D \mathbb{E}(X|\mathcal{G}) d\,\mathbb{P}, \quad D \in \mathcal{G}.$$
   So $\mathbb{E}(X|\mathcal{G})$ is a $\mathcal{G}$-measurable function whose integral on any element of $\mathcal{G}$ is greater or equal to 0. This implies that it must be positive a.s.

3. Let $D \in \mathcal{G}$
$$\begin{aligned}\int_D E(aX_1 + bX_2|\mathcal{G}) d\,\mathbb{P} &= \int_D (aX_1 + bX_2) d\,\mathbb{P} \\ &= a\int_D X_1 d\,\mathbb{P} + b\int_D X_2 d\,\mathbb{P} \\ &= a\int_D E(X_1|\mathcal{G}) d\,\mathbb{P} + b\int_D \mathbb{E}(X_2|\mathcal{G}) d\,\mathbb{P} \\ &= \int_D [aE(X_1|\mathcal{G}) + bE(X_2|\mathcal{G})]\, d\,\mathbb{P}\end{aligned}$$

4. It is an immediate consequence of 1.9-(2) and 1.9-(3)

5. By the property of monotony (4) we get that $\{\mathbb{E}(X_n|\mathcal{G})\}_{n=1}^{\infty}$ is non decreasing.
   Let $Z = \lim_{n\to\infty} \mathbb{E}(X_n|\mathcal{G})$ and $D \in \mathcal{G}$. By the usual monotone theorem first applied to $\{\mathbb{1}_D X_n\}_{n\in\mathbb{N}}$:
$$\lim_{n\to\infty} \int_D X_n d\,\mathbb{P} = \int_D \lim_{n\to\infty} X_n d\,\mathbb{P} = \int_D X d\,\mathbb{P} = \int_D \mathbb{E}(X|\mathcal{G}) d\,\mathbb{P}$$
   and then applied to $(\mathbb{1}_D\, \mathbb{E}(X_n|\mathcal{G}))_n$
$$\lim_{n\to\infty} \int_D X_n d\,\mathbb{P} = \lim_{n\to\infty} \int_D \mathbb{E}(X_n|\mathcal{G}) d\,\mathbb{P} = \int_D \lim_{n\to\infty} \mathbb{E}(X_n|\mathcal{G}) d\,\mathbb{P} = \int_D Z d\,\mathbb{P}$$
   we obtain $\int_D Z d\,\mathbb{P} = \int_D \mathbb{E}(X|\mathcal{G}) d\,\mathbb{P}$ for all $D \in \mathcal{G}$: This means that $Z = \mathbb{E}(X|\mathcal{G})$ a.s.

6. Introduce the auxiliary functions $U_n := \inf_{m\geq n} X_m \leq X_n \leq \sup_{m\geq n} X_m := V_n$, then

   ♢ $-Y \leq U_n \leq U_{n+1} \leq X_{n+1}$ and

   ♢ $\lim_{n\to\infty} U_n = \liminf_{n\to\infty} X_n = \lim_{n\to\infty} X_n = X$.

   So we can apply the CMCT (5) to the increasing sequence $\{U_n + Y\}_{n\in\mathbb{N}}$ of positive functions and obtain
   $$\lim_{n\to\infty} \mathbb{E}(U_n + Y|\mathcal{G}) = \mathbb{E}(X + Y|\mathcal{G}),$$
   so,
   $$\mathbb{E}(X|\mathcal{G}) = \lim_{n\to\infty} \mathbb{E}(U_n|\mathcal{G}) \tag{1.1}$$
   since $\mathbb{E}(Y|\mathcal{G})$ is finite a.s.

   We show in a similar way, using fact that $\{Y - V_n\}_{n\in\mathbb{N}}$ is again increasing sequence of positive functions, that
   $$\lim_{n\to\infty} \mathbb{E}(Y - V_n|\mathcal{G}) = \mathbb{E}(Y - X|\mathcal{G}),$$
   so $\lim_{n\to\infty} \mathbb{E}(V_n|\mathcal{G}) = \mathbb{E}(X|\mathcal{G})$.

   Using monotony (4) we get
   $$\begin{aligned}\mathbb{E}(X|\mathcal{G}) &= \lim_{n\to\infty} \mathbb{E}(V_n|\mathcal{G}) \\ &\geq \lim_{n\to\infty} \mathbb{E}(X_n|\mathcal{G}) \\ &\geq \lim_{n\to\infty} \mathbb{E}(U_n|\mathcal{G}) \\ &= \mathbb{E}(X|\mathcal{G}).\end{aligned}$$
   This implies the desired result $\mathbb{E}(X|\mathcal{G}) = \lim_{n\to\infty} \mathbb{E}(X|\mathcal{G})$

7. We using similar arguments, define the functions $U_n := \inf_{m\geq n} X_m \leq X_n$, then

   ♢ $0 \leq U_n \leq U_{n+1}$ for every $n \in \mathbb{N}$.

   ♢ $\lim_{n\to\infty} \mathbb{E}(U_n|\mathcal{G}) = \mathbb{E}(\lim_{n\to\infty} U_n|\mathcal{G}) = \mathbb{E}(\liminf_{n\to\infty} X_n|\mathcal{G})$ (we used CMCT).

   ♢ $\lim_{n\to\infty} \mathbb{E}(U_n|\mathcal{G}) = \liminf_{n\to\infty} \mathbb{E}(U_n|\mathcal{G}) \leq \liminf_{n\to\infty} \mathbb{E}(X_n|\mathcal{G})$.

   These last two facts prove the lemma.

8. Let $g\colon \mathbb{R} \to \mathbb{R}$ be a convex function such that $g(X) \in L_1(\Omega, \mathcal{F}, \mathbb{P})$. A well known result on convex functions says that for all $x \in \mathbb{R}$
   $$g(x) = \sup\{ax + b\colon (a, b) \in S\}$$
   where $S = \{(a, b)\colon ax + b \leq g(x)\}$. Then for all $(a, b) \in S$
   $$\begin{aligned}\mathbb{E}(g(X)|\mathcal{G}) &\geq \mathbb{E}(aX + b|\mathcal{G}) \\ &= a\,\mathbb{E}(X|\mathcal{G}) + b\end{aligned}$$
   Taking the supremum over all $(a, b) \in S$ we obtain the result.

■

Observation 1.10. *The same theorem holds true for the evaluated conditional expectation. For instance under the hypoteses in 3 of the former theorem, if $\mathcal{G} = \sigma(Y)$ for some random variable $Y\colon (\Omega, \mathcal{F}, \mathbb{P}) \to \mathbb{R}$ we can write for any $y \in \mathbb{R}$,*

$$\mathbb{E}(X|Y = y) = \lim_{n\to\infty} \mathbb{E}(X_n|Y = y) \quad P_Y\text{-a.s}$$

where P_Y is the law of Y.

1.3 Second group of properties.

This group of properties is related to measurability and expectation.

Proposition 1.11. *Let $X, X_1 \in L_1(\Omega, \mathcal{F}, \mathbb{P})$, Y be a random variable in $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G}$ a sub–σ-algebra of $\mathcal{F}$. then*

1. *If X is $\mathcal{G}$-measurable then $\mathbb{E}(X|\mathcal{G}) = X$.*
2. *$\mathbb{E}(\mathbb{E}(X|\mathcal{G})) = \mathbb{E}(X)$ and $\int_{\mathbb{R}} \mathbb{E}(X|Y = y)dP_Y(y) = \mathbb{E}(X)$*
   where P_Y is the law of Y and $\mathbb{E}(X|Y = y)$ is the evaluated conditional expectation (see the definitions 1.4 and 1.8 of section 1.)
3. *If $\mathcal{G} = \{\emptyset, \Omega\}$ then $\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X)$.*
4. *If X is $\mathcal{G}$-measurable $\mathbb{E}(X_1X|\mathcal{G}) = X\,\mathbb{E}(X_1|\mathcal{G})$ provided $X_1X \in L_1(\Omega, \mathcal{F}, \mathbb{P})$.*
5. *If $\mathcal{A}$ is a sub-σ-algebra of $\mathcal{G}$,*
$$\mathbb{E}(X|\mathcal{A}) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{A}) = \mathbb{E}(\mathbb{E}(X|\mathcal{A})|\mathcal{G})$$

Proof.

1. Immediate from the definition and it means that since X is already $\mathcal{G}$-measurable, there is nothing to be changed in X by conditioning with $\mathcal{G}$.
2. The first part is obtained if we apply the definition of conditional expectation to the set $\Omega \in \mathcal{G}$:
$$\mathbb{E}(X) = \int_\Omega X d\,\mathbb{P} = \int_\Omega Z d\,\mathbb{P} = \mathbb{E}(\mathbb{E}(X|\mathcal{G})).$$
   The second part is an easy consequence of the change of variable formula.
3. It is immediate if we recall that the only measurable functions with respect to the trivial σ-algebra $\mathcal{G} = \{\emptyset, \Omega\}$ are the constant functions.
4. We first suppose that $X = \mathbb{1}_F$ and $F \in \mathcal{G}$. If $D \in \mathcal{G}$, we have in fact $F, D, (F \cap D) \in \mathcal{G}$ so by definition of conditional expectation we have
$$\begin{aligned}\int_D \mathbb{E}(\mathbb{1}_F X_1|\mathcal{G})d\,\mathbb{P} &= \int_D \mathbb{1}_F X_1 d\,\mathbb{P}\\ &= \int_{D\cap F} X_1 d\,\mathbb{P}\\ &= \int_{D\cap F} \mathbb{E}(X_1|\mathcal{G})d\,\mathbb{P}\\ &= \int_D \mathbb{1}_F\,\mathbb{E}(X|\mathcal{G})d\,\mathbb{P}\,.\end{aligned}$$

This means that the result holds true for $X = \mathbb{1}_F$.

Now let X be a positive and simple $\mathcal{F}$ measurable function, by linearity the result holds for this X. For $X\mathcal{F}$ measurable and positive , let $(Z_n)_{n\in\mathbb{N}}$ a non decreasing sequence of positive simple measurable functions that approximate X. The result just proved for simple functions allows us to write for all $n \in \mathbb{N}$,

$$\mathbb{E}(X_1 Z_n|\mathcal{G}) = Z_n\,\mathbb{E}(X_1|\mathcal{G}),$$

Since $|Z_n X_1| \leq Z_n|X_1| \leq X|X_1| \in L_1\,(\Omega, \mathcal{F}, \mathbb{P})$ we can apply the CDCT to conclude.

Finally for X-measurable we use the usual decomposition $X = X^+ - X^-$ and apply the former result to each term. Observe that X^+X_1 and (X^-X_1) both belong to $L_1\,(\Omega, \mathcal{F}, \mathbb{P})$.

5. The equality $\mathbb{E}(X|\mathcal{A}) = \mathbb{E}(\mathbb{E}(X|\mathcal{A})|\mathcal{G})$ is inmediate from Proposition 1.11-(1) since $\mathbb{E}(X|\mathcal{A})$ is $\mathcal{G}$-measurable.

   To see the other equality $\mathbb{E}(X|\mathcal{A}) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{A})$, we need to prove for each $A \in \mathcal{A}$ the equality:

$$\int_A X d\,\mathbb{P} = \int_A \mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{A}) d\,\mathbb{P}$$

   But this is true since,

$$\int_A \mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{A}) d\,\mathbb{P} = \int_A \mathbb{E}(X|\mathcal{G}) d\,\mathbb{P} = \int_A X d\,\mathbb{P}$$

■

1.4 Relationship with independence

Proposition 1.12. *Let $\mathcal{G}, \mathcal{A}$ be sub-σ-algebras of $\mathcal{F}$ and $X \in L_1(\Omega, \mathcal{F}, \mathbb{P})$.*

1. *If $\sigma(X)$ and $\mathcal{G}$ are independent, then*

$$\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X)$$

   In particular if $Y : \Omega \to \mathbb{R}$ is another measurable function and $\sigma(X)$ is independent of $\sigma(Y)$ then

$$\mathbb{E}(X|\sigma(Y)) = \mathbb{E}(X)$$

2. * *If $H : \mathbb{R}^2 \to \mathbb{R}$ is Borel measurable and bounded and X, Y are independent random variables*

$$\mathbb{E}(H(X,Y)|\sigma(Y)) = \int_{\mathbb{R}} H(x,Y) dP_X(x), \tag{1.2}$$

   (recall the definition of the the law P_X of X in 1.4 of section 1.)

3. * *If $\sigma(X)$ and $\mathcal{A}$ are independent of $\mathcal{G}$ then*

$$\mathbb{E}(X|\sigma(\mathcal{A} \cap \mathcal{G})) = \mathbb{E}(X|\mathcal{A})$$

   In particular, if $Y, Z : \Omega \to \mathbb{R}$ are measurable functions such that X y Z are independent of Y then

$$\mathbb{E}(X|\sigma(Y,Z)) = \mathbb{E}(X|\sigma(Z)).$$

Proof.

1. If $D \in \mathcal{G}$, the independence implies

$$\int_D X d\,\mathbb{P} = \mathbb{P}(D) \cdot \mathbb{E}(X) = \int_D \mathbb{E}(X) d\,\mathbb{P}$$

and this proves the first part of the assertion. The second one follows form this one.

2. We suppose that $H(x,y) = \mathbb{1}_A(x)\mathbb{1}_B(y) = \mathbb{1}_{A\times B}(x,y)$ where A, B are any sets in $\mathcal{B}(\mathbb{R})$. The result is, in this case, immediate because we can write,

$$\begin{aligned}
\mathbb{E}(H(X,Y)|\sigma(Y)) &= \mathbb{E}(\mathbb{1}_A(X)\mathbb{1}_B(Y)|\sigma(Y)) \\
&= \mathbb{1}_B(Y)\,\mathbb{E}(\mathbb{1}_A(X)|\sigma(Y)) \\
&= \mathbb{1}_B(Y)\,\mathbb{E}(\mathbb{1}_A(X)) \\
&= \mathbb{1}_B(Y)\int_{\mathbb{R}} \mathbb{1}_A(x) dP_X(x) \\
&= \int_{\mathbb{R}} H(x,Y) dP_X(x)
\end{aligned}$$

The second equality comes from the fact that $\mathbb{1}_B(Y)$ is $\sigma(Y)$-measurable and the fourth one is just the change of variable formula applied to $f(x) = \mathbb{1}_A(x)$. Now we take the family

$$\mathcal{L} = \{C \in \mathcal{B}(\mathbb{R}^2)\colon \text{ formula (1.2) holds true for } H(x,y) = \mathbb{1}_C(x,y)\}.$$

Since we know that the family $\mathcal{C} = \{A \times B, A, B \in \mathcal{B}(\mathbb{R})\}$ is stable under finite intersection, generates $\mathcal{B}(\mathbb{R}^2)$ and is contained in $\mathcal{L}$ (first part of this proof), all we need to prove, thanks to the monotone class lemma, is that the family $\mathcal{L}$ is a λ-system. To see this, let

- ♢ $C_1, C_2 \in \mathcal{L}$ with $C_1 \subset C_2$. We then have

$$\mathbb{E}(\mathbb{1}_{C_i}(X,Y)|\sigma(Y)) = \int_{\mathbb{R}} \mathbb{1}_{C_i}(x,Y) dP_X(x), \quad i = 1,2$$

  Since $\mathbb{1}_{C_2 - C_1} = \mathbb{1}_{C_2} - \mathbb{1}_{C_1}$ the conclusion $(C_2 - C_1) \in \mathcal{L}$ follows.
- ♢ $C_n \in \mathcal{L}, \;\; C_{n-1} \subset C_n, \;\; n = 1, 2, \ldots,$ and $C = \cup_{n\in\mathbb{N}} C_n$.

$$\mathbb{E}(\mathbb{1}_{C_n}(X,Y)|\sigma(Y)) = \int_{\mathbb{R}} \mathbb{1}_{C_n}(x,Y) dP_X(x), \quad n = 1, 2, \ldots$$

The desired result follows from the monotone convergence theorem applied to both sides of the equality because,

- ♢ $\mathbb{1}_{C_{n-1}} \leq \mathbb{1}_{C_n}$, and
- ♢ $\mathbb{1}_C = \lim_{n\to\infty} \mathbb{1}_{C_i}$.

3. If $D \in \sigma(Y)$

$$\begin{aligned}
\int_D \mathbb{E}(X|Y) d\,\mathbb{P} &= \int_D X d\,\mathbb{P} = \int_\Omega \mathbb{1}_D X d\,\mathbb{P} \\
&= \mathbb{E}(\mathbb{1}_D)\,\mathbb{E}(X) = \int_D \mathbb{E}(X) d\,\mathbb{P}
\end{aligned}$$

Let $\mathcal{C} = \{A \cup D : A \in \mathcal{A},\ D \in \mathcal{G}\}$ and $\mathcal{L} = \{D \in \mathcal{F} | \int_D X d\,\mathbb{P} = \int_D \mathbb{E}(X|\mathcal{A}) d\,\mathbb{P}\}$
We must show that for any $F \in \sigma(A \cup D)$ we have:

$$\int_F X d\,\mathbb{P} = \int_F \mathbb{E}(X|\mathcal{A}) d\,\mathbb{P}$$

For this it is enough to prove that $\mathcal{C}$ is stable under finite intersections and that $\mathcal{L}$ is a λ-system such that $\mathcal{C} \subset \mathcal{L}$. The first two assertions are easy to prove, to see the third one, let $A \in \mathcal{A}$ and $D \in \mathcal{G}$:

$$\begin{aligned}
\int_{(D\cup A)} X d\,\mathbb{P} &= \int_\Omega \mathbb{1}_{D\cup A} X d\,\mathbb{P} \\
&= \int_\Omega (\mathbb{1}_D + \mathbb{1}_A - \mathbb{1}_{D\cap A}) X d\,\mathbb{P} \\
&= \mathbb{E}(\mathbb{1}_D)\,\mathbb{E}(X) + \mathbb{E}(\mathbb{1}_A X) - \mathbb{E}(\mathbb{1}_D)\,\mathbb{E}(\mathbb{1}_A X) \\
&= \mathbb{E}(\mathbb{1}_D) \int_\Omega \mathbb{E}(X|\mathcal{A}) d\,\mathbb{P} + \int_A \mathbb{E}(X|\mathcal{A}) d\,\mathbb{P}\,| + \mathbb{E}(\mathbb{1}_D) \int_A \mathbb{E}(X|\mathcal{A}) d\,\mathbb{P} \\
&= \int_\Omega \mathbb{1}_D\,\mathbb{E}(X|\mathcal{A}) d\,\mathbb{P} + \int_A \mathbb{E}(X|\mathcal{A}) d\,\mathbb{P} - \int_A \mathbb{1}_D\,\mathbb{E}(X|\mathcal{A}) d\,\mathbb{P} \\
&= \int_\Omega (\mathbb{1}_D + \mathbb{1}_A - \mathbb{1}_{D\cap A})\,\mathbb{E}(X|\mathcal{A}) d\,\mathbb{P} \\
&= \int_{D\cap A} \mathbb{E}(X|\mathcal{A}) d\,\mathbb{P}
\end{aligned}$$

We used the definition of conditional expectation and the given hypotheses of independence. Notice that in order to have

$$\int_\Omega \mathbb{1}_{D\cap A} X d\,\mathbb{P} = \mathbb{E}(\mathbb{1}_D) \int_\Omega \mathbb{1}_A X d\,\mathbb{P}$$

and

$$\mathbb{E}(\mathbb{1}_D) \int_A \mathbb{E}(X|\mathcal{A}) d\,\mathbb{P} = \int_A \mathbb{1}_D\,\mathbb{E}(X|\mathcal{A}) d\,\mathbb{P}$$

both independence hypotheses are used: $\mathcal{G}$ is independent of X and of $\mathcal{A}$.

■

1.5 Exercises

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space in which $\mathcal{G}$ is a sub-σ-algebra of $\mathcal{F}$. The equalities must be seen as valid almost surely.

1. Let $X : \Omega \to \mathbb{R}$ a positive $\mathcal{F}$-measurable function. If $Q(A) := \int_A X d\mathbb{P}$ where $A \in \mathcal{F}$, then
   (a) Q is a measure on $(\Omega, \mathcal{F})$.
   (b) $Q << \mathbb{P}$ i.e. if $\mathbb{P}(B) = 0$, then $Q(B) = 0$.
2. (a) Let $B \in \mathcal{F}$ be such that $\mathbb{P}(B) > 0$. Define $Q^B : \mathcal{F} \to [0, 1]$ as
   $$Q^B(A) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad A \in \mathcal{F}$$
   Show that Q^B is a probability measure in $\mathcal{F}$ usually denoted by $P(\cdot|B)$ and it is called the conditional probability given $B \in \mathcal{F}$ (this is the elementary form of conditional expectation studied in the basic probability courses).
   (b) If $X \in L_1(\Omega, \mathcal{F}, P)$ then $X \in L_1(\Omega, \mathcal{F}, Q^B)$ and the **conditional expectation of** X **given** B is defined as
   $$\mathbb{E}(X|B) := \int_\Omega X dQ^B = \frac{1}{\mathbb{P}(B)} \int_B X d\mathbb{P} = \frac{\mathbb{E}(\mathbb{1}_B X)}{\mathbb{P}(B)}.$$
   Observe that here $\mathbb{E}(X|B)$ this is just a number, not a random variable.
3. Find a sequence of random variables $X_n \in L_1(\Omega, \mathcal{F}, \mathbb{P}), \quad n \in \mathbb{N}$ which is not dominated by an integrable random variable : there does not exist a $Y \in L_1$ that verifies $|X_n| \leq |Y|$ for all $n \in \mathbb{N}$.
4. If X, Y are random variables $\mathcal{G}$-measure and $\int_D d\mathbb{P} = \int_D Y d\mathbb{P}$ for all $D \in \mathcal{G}$, then $X = Y$ a.s.
5. If $\mathcal{A}$ is the σ-algebra generated by $B \in \mathcal{F}$ with $P(B) > 0$ y $P(B^c) > 0$ find using the former exercise,
   (a) $\mathbb{E}(X|\mathcal{A})$ for $X \in L_1(\Omega, \mathcal{F}, \mathbb{P})$.
   (b) $\mathbb{P}(C|\mathcal{A})$ for $C \in \mathcal{F}$.
6. (a) If $\mathcal{A}$ is the σ-algebra generated by a measurable partition of Ω, $B_1, B_2, \ldots, B_n$ such that $P(B_i) > 0$, $i = 1, 2, \ldots, n$ and if $X \in L_1(\Omega, \mathcal{F}, \mathbb{P})$ find $Y = \mathbb{E}(X|\mathcal{A})$.
   (b) One important particular case is the following: let $(\Omega', \mathcal{F}')$ be another measurable space and Y a measurable function $Y : (\Omega, \mathcal{F}) \to (\Omega', \mathcal{F}')$. If $\operatorname{Im} Y = \{y_1, y_2, \ldots, y_n\} \subset \Omega'$ with $\mathbb{P}(Y = y_i) > 0$, $i = 1, 2, \ldots, n$ then we have a measurable partition of Ω given by $\{B_i = (Y = y_i) : i = 1, 2, \ldots, n\}$: If $\mathcal{A} = \sigma(Y)$ find $\mathbb{E}(X|\sigma(Y))$ and $\mathbb{E}(X|Y = y_i)$, $i = 1, 2, \ldots, n$.

7. Another important concept from the basic courses is the following: the random vector (X, Y) has a joint density function $f\colon \mathbb{R}^2 \to [0,1]$. Then the definition of **the conditional density** of X given Y is defined as

$$h(x|y) := \frac{f(x,y)}{f_Y(y)}$$

where f_Y is the marginal of Y and it is supposed that $f_Y(y) > 0$, $y \in \mathbb{R}$.
The following definition is given

$$\mathbb{E}(X|Y=y) = \int_{\mathbb{R}} x h(x|y) dx = \frac{1}{f_Y(y)} \int_{\mathbb{R}} x f(x,y) dx.$$

and also

$$\mathbb{E}(v(X)|Y=y) = \int_{\mathbb{R}} v(x) h(x|y) dx = \frac{1}{f_Y(y)} \int_{\mathbb{R}} v(x) f(x,y) dx.$$

for any continuous and bounded $v\colon \mathbb{R} \to \mathbb{R}$.

Prove that this coincides with the evaluated conditional expectation $\mathbb{E}(v(X)|Y = y)$ given in definition 1.8 that this gives us a way to obtain explicitly

$$\mathbb{E}(v(X)|\sigma(Y)).$$

Also if

$$P(X \in A|Y=y) := \int_A h(x|y) dx = \frac{1}{f_Y(y)} \int_A f(x,y) dx$$

What can you say of the relation of this definition to our definitions.

8. Prove the first part of proposition 1.5.

9. Prove that $\mathcal{B}(\mathbb{R}^2)$ is equal to the σ-algebra generated by the family

$$\mathcal{C} = \{A \times B\colon A, B \in \mathcal{B}(\mathbb{R})\}.$$

10. Suppose that $\Omega = [-1,1]$, $\mathcal{F} = \mathcal{B}(\mathbb{R}), \mathbb{P} = \frac{\lambda}{2}$, where λ is the Lebesgue measure in $[-1,1]$. If $X, X_i : \Omega \to \mathbb{R}, i = 1,2,3$ are defined as

    ♢ $X(\omega) = \omega$,
    ♢ $X_1(\omega) = |\omega|$,
    ♢ $X_2(\omega) = 2\omega$,
    ♢ $X_3(\omega) = \omega^2$.

    Give the explicit form of

    (a) $\mathbb{E}(X|X_i)$, $i = 1,2,3$,
    (b) $\mathbb{E}(X_i|X)$, $i = 1,2,3$,
    (c) $P(X > 0|X_1)$.

1.6 Appendix

Finally we will prove a technical but very useful result called **monotone class lemma**. This has been used in the last part of the properties of conditional expectation and is, in general, very useful. A class $\mathcal{A}$ of subsets of a fixed space Ω is called

- ♢ stable under finite intersections if whenever $A, B \in \mathcal{A}$ we also have $A \cap B \in \mathcal{A}$.
- ♢ stable under increasing limits if $C_n \in \mathcal{A}$, $C_{n-1} \subset C_n$, $n = 1, 2, \ldots$ and $C = \cup_{n\in\mathbb{N}} C_n$ then $C \in \mathcal{A}$.
- ♢ stable under proper differences if $C_1, C_2 \in \mathcal{A}$ with $C_1 \subset C_2$implies that $(C_2 - C_1) \in \mathcal{A}$
- ♢ the class $\mathcal{A}$ is called λ-system if it is stable under proper differences and increasing limits.

Lemma 1.13. *Monotone class lemma. Let $\mathcal{C}$ be a class of subsets of Ω stable under finite intersections and containing Ω. Let $\mathcal{L}$ be the smallest λ-system in Ω such that*

$$\mathcal{C} \subset \mathcal{L}.$$

Then $\mathcal{L} = \sigma(\mathcal{C})$ i.e. $\mathcal{L}$ is the σ-algebra generated by $\mathcal{C}$.

Proof. Observe that any intersection of λ- systems is again a λ- system, so the smallest λ-system that contains $\mathcal{C}$ is a well defined family of subsets of Ω.

For any $B \subset \Omega$ we define

$$\mathcal{D}_B = \{A \subset \Omega \colon A \in \mathcal{L} \text{ and } A \cap B \in \mathcal{L}\}.$$

It is easy to verify that $\mathcal{D}_B$ is a λ-system.

Now take $B \in \mathcal{C}$. For each $C \in \mathcal{C}$ we have $B \cap C \in \mathcal{C} \subset \mathcal{L}$ and we get $C \in \mathcal{D}_B$. This implies

$$\mathcal{C} \subset \mathcal{D}_B \subset \mathcal{L}.$$

But since $\mathcal{D}_B$ is a λ-system containing $\mathcal{C}$ we conclude that

$$\mathcal{L} = \mathcal{D}_B.$$

Now let $B \in \mathcal{L}$. The former equality allow us to say that for all $C \in \mathcal{C}$ we have $B \in \mathcal{D}_C$ so $B \Cap C \in \mathcal{L}$, hence $\mathcal{L}$ is stable under finite intersections. Additionally $\Omega \in \mathcal{C} \subset of\mathcal{L}$, so complements are in $\mathcal{L}$. The proof is done since under these conditions $\mathcal{L}$ is a σ-algebra .

Chapter 2

Stochastic processes

Definition 2.1 (Stochastic process)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(S, \mathcal{A})$ a measurable space. A stochastic process with values in the state space $(S, \mathcal{A})$ is a family of random variables $X = \{X_t : t \in J\}$, defined in $(\Omega, \mathcal{F}, \mathbb{P})$ and with values in S. Usually J (the time parameter) is*

1. *a finite or infinite subset of $\mathbb{N}$, and we say that the process is* **discrete in time**.

2. *an interval of $\mathbb{R}^+$ of the form $[0, \infty)$ or $[a, b]$, with $0 \leq a < b$ and we say it is* **continuous in time.**

The most frequent case in this course will be $S = \mathbb{R}$ and $\mathcal{A} = \mathcal{B}(\mathbb{R})$ i.e. the Borel σ-algebra in $\mathbb{R}$ and $J = [0, \infty)$ or $J = [0, T]$. For each $\omega \in \Omega$ fixed, the function $t \to X_t(\omega)$ is called a **path** of the process X. We can also regard a process as a function defined on the product space

$$X : J \times \Omega \to S$$

and write $X(t, \omega)$ instead of $X_t(\omega)$. When J is an interval of $\mathbb{R}$, we say that: the process is a.s. *right continuous* (resp. *left continuous*) if for almost all $\omega \in \Omega$ the path $t \to X_t(\omega)$ is right continuous, (resp. left continuous). A process is *continuous* if for almost all $\omega \in \Omega$ the path $t \to X_t(\omega)$ is continuous.

Definition 2.2. *There are several forms of equality amongst processes; for two processes X and Y defined on the same probability space we say that*

1. *X is a* **modification** *of Y if for all $t \in J$, $\mathbb{P}(X_t = Y_t) = 1$*

2. *X is* **indistinguishable** *from Y if $\mathbb{P}(X_t = Y_t,\ \text{for all}\ t \in J) = 1$, this means that there exists a set $N \in \mathcal{F}$ with $\mathbb{P}(N) = 0$ and for each $\omega \notin N$ we have that $X_t(\omega) = Y_t(\omega)$, for all $t \in J$. This means that a.s. they have the same paths.*

We can see that even if two processes are a modification of each other, they might have different paths (exercise). Clearly if X is indistinguishable from Y, then X is a modification of Y. If we have some regularity of the paths, the both notions are equivalent:

Proposition 2.3. *If X, Y are right continuous (or left continuous) and X is a modification of Y, then they are indistinguishable.*

Proof. Let D be a countable dense subset of $J = [0, \infty)$ and N the complement of the set $\{X_t = Y_t, \text{for all } t \in D\}$. Then because of the countable sub-additivity of $\mathbb{P}$ we have

$$\mathbb{P}(N) \leq \sum_{t \in D} \mathbb{P}(X_t \neq Y_t) = 0$$

Let $C \in \mathcal{F}$ be the set where the paths of X and Y are right continuous. Then $\mathbb{P}(C) = 1$ and if $A := N \cup C^c$ then

$$\mathbb{P}(A) = \mathbb{P}(N \cup C^c) = 0.$$

The set $A^c = N^c \cap C$ has probability 1 and by the right continuity of the paths and the density of D we have $X_t = Y_t, \text{for all } t \in J$, so $\mathbb{P}(X_t = Y_t, \text{for all } t \in J) = \mathbb{P}(A^c) = 1$. ■

Definition 2.4. *The* **finite dimensional distributions** *of a process are*

$$\mathbb{P}(X_{t_1} \in B_1, X_{t_2} \in B_2, \ldots, X_{t_k} \in B_k)$$

with $k \in \mathbb{N}$, $t_i \in J$, $B_i \in \mathcal{B}(\mathbb{R})$, $i = 1, 2, \ldots, k$.

They are important, because in many occasions this is all we know about the process, and they are essential to construct a process (Kolmogorov consistency theorem).

It is clear that: if X is a modification or a version of Y then they have the same finite dimensional distributions (exercise).

Definition 2.5. *Let X, Y processes (that might be defined in different probability spaces), they are said to be* **equivalent** *if they have the same finite dimensional distributions.*

Definition 2.6. *A* **filtration** $\{\mathcal{F}_t : t \in J\}$ *or* $(\mathcal{F}_t)_{t \in J}$ *in* $(\Omega, \mathcal{F}, \mathbb{P})$ *is a family of sub-σ-algebras of $\mathcal{F}$ such that $\mathcal{F}_s \subset \mathcal{F}_t$, $s < t$, $s, t \in J$. The probability space together with the filtration is denoted* $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in J}, \mathbb{P})$.

Definition 2.7. *Let $\{X_t : t \in J\}$ a stochastic process, we say that X is*

1. **measurable** *if $X : J \times \Omega \to S$ is $\mathcal{B}(J) \otimes \mathcal{F}$-measurable.*
2. **adapted** *to the filtration $\{\mathcal{F}_t : t \in J\}$ if for all $t \in J$, X_t is $\mathcal{F}_t$-measurable.*
3. **progressively measurable** *or simply* **progressive** *if for all $t > 0$ the function $(s, \omega) \to S$ from $[0, t] \times \Omega$ to S is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$-measurable.*

These definitions are interesting in the continuous time setting. In the discrete time case, all we usually need is that the process is adapted in the space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n \in J}, \mathbb{P})$.

Definition 2.8. *The* **canonical filtration** *of a given process $\{X_t : t \in J\}$ is the filtration generated by the process, i.e. for each $t \in J$, $\mathcal{F}_t$ is the σ-algebra generated by the family of random variables $\{X_s : s \leq t, s \in J\}$, which can also be written as*

$$(\mathcal{F}_t = \sigma(X_s : s \in J, s \leq t), \quad t \in J).$$

It is obvious that any process is adapted to its canonical filtration.

Proposition 2.9. *A right (or left) continuous process is measurable.*

Proof. We will do it in the case $J = [0, \infty)$. By definition of the product σ-algebra, for each $n \in \mathbb{N}$, the mapping $\phi_n : J \times \Omega \to \mathbb{R}$ defined as

$$\Phi_n(t, \omega) = X_{\frac{[2^n t]}{2^n}}(\omega)$$

is $\mathcal{B}([0, \infty)) \otimes \mathcal{F}$-measurable for all $n \in \mathbb{N}$. Then by right continuity we conclude since s $\frac{[2^n t]}{2^n} \geq t$ and by right continuity

$$\lim_{n \to \infty} \Phi_n(t, \omega) = X_t(\omega)$$

(recall that the limit of measurable functions is measurable). ■

The same idea is used to prove the following result and it is left as an exercise.

Proposition 2.10. *A right (or left) continuous adapted process is progressive.*

Definition 2.11. *We will need several concepts related to the filtration:*

A **discrete stopping time** *in* $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{n}})_{\mathrm{n} \in \mathrm{J}}, \mathbb{P})$ *is a random variable*

$$\tau : \Omega \to \mathbb{N} \cup \{+\infty\}$$

such that $\{\tau \leq n\} \in \mathcal{F}_n$ *for all* $n \in J \subset \mathbb{N}$. *It is immediate that* $\tau : \Omega \to \mathbb{N} \cup \{+\infty\}$ *is a discrete stopping time iff* $\{\tau = n\} \in \mathcal{F}_n$ *for all* $n \in J \subset \mathbb{N}$

A **stopping time**, *with respect to given filtration* $\{\mathcal{F}_t : t \in J\}$ *is a random variable* $\tau : \Omega \to [0, \infty]$ *such that* $\{\tau \leq t\} \in \mathcal{F}_t$ *for all* $t \in J$.

Given a stopping time τ *and a process* X, *the mapping* $X_\tau : \Omega \to S$ *is defined as*

$$X_\tau(\omega) := X_{\tau(\omega)}(\omega).$$

Remark 2.12. *If* X *is measurable and* τ *is a finite stopping time, then* X_τ *is a random variable (exercise).*

Examples 2.13.

1.- Random Walk (RW): if $(\xi_n)_{n \in \mathbb{N}}$ is a sequence of independent identically distributed random variables, the corresponding **random walk** is a discrete time process X defined as:

$$X_n := a + \sum_{i=1}^{n} \xi_i, \quad a \in \mathbb{R}^+ .$$

We define,

$$\tau = \begin{cases} \min\{n \in \mathbb{N} : X_n \geq 2a\}, & \text{if } \{n \in \mathbb{N} : X_n \geq 2a\} \neq \emptyset, \\ \infty \text{ if } \{n \in \mathbb{N} : X_n \geq 2a\} = \emptyset \end{cases}$$

Then τ is a discrete stopping time w.r.t the canonical filtration of the RW, since for all $m \in \mathbb{N}$, $\{\tau \leq m\} = \bigcup_{i=1}^{m} \{X_i \geq 2a\} \in \mathcal{F}_m$.

2.- $\tau = \max\{n \in \mathbb{N} : X_n \geq 2a\}$ is not a stopping time.

3.- In fact, a similar result is true if we replace the R.W. by any adapted discrete time process.

4.- If $X = (X_t : t \geq 0)$ is a process with continuous paths and A is a closed subset of $\mathbb{R}$ then **the hitting time** of a set A is defined as:

$$H_A = \begin{cases} \inf\{t \geq 0 : X_t \in A\}, & \text{if } \{t \geq 0 : X_t \in A\} \neq \emptyset, \\ \infty, & \text{if } \{t \geq 0 : X_t \in A\} = \emptyset \end{cases} \tag{2.1}$$

is a stopping time. This is left as an exercise.

Proposition 2.14. *For any stopping time $\tau : \Omega \to [0, \infty]$ there exists a decreasing sequence $\tau_n : \Omega \to [0, \infty]$ $n = 1, 2, 3, \ldots$ of discrete stopping times , such that*

$$\tau(\omega) = \lim_{n \to \infty} \tau_n(\omega)$$

Proof. We define

$$\tau_n(\omega) = \begin{cases} \frac{q}{2^n} & \text{if } (q-1)2^{-n} \leq \tau(\omega) < q2^{-n}, \\ \infty & \text{if } \tau(\omega) \geq n \end{cases}$$

It is easily seen that τ_n is a stopping time and that τ_n decreases to s τ.

Observation: Use and prove (see 8.- in exercises 2.1) that if τ is a stopping time then for all $t \geq 0$, we have that $(\tau < t) \in \mathcal{F}_t$.

2.1 Exercises

(2.1)

Here we write $\mathbb{N}$ for the set $\{1, 2, 3, ...\}$. According to each exercise we are considering the canonical filtration or else we work in a fixed space $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in \mathrm{J}}, \mathbb{P})$.

1. Give an example of two stochastic processes that are modification of each other and but are not indistinguishable.
2. If X is indistinguishable from Y, then X is a modification of Y.
3. If X is a modification or a version of Y then they have the same finite dimensional distribution.
4. Prove that right (or left) continuous adapted process is progressive.
5. If τ is constant, then it is a stopping time.
6. If τ is a stopping time and $x \in \mathbb{R}$, is it true that $\tau + x$ is a stopping time?
7. Simple random walk (SRW): if $(\xi_n)_{n\in\mathbb{N}}$ is a sequence of independent identically distributed Bernoulli with parameter $p \in (0, 1)$ and values in the set $-1, 1$, the corresponding **random walk** is a discrete time process X defined as: $X_n := a + \sum_{i=1}^{n} \xi_i, \;\; a \in \mathbb{R}$. Compute the distribution of the stopping time defined by
$$\tau_0 = \min\{n \in \mathbb{N} \colon X_n = 0\}, \;\; \text{if} \;\; a = 0$$
8. Prove that if τ is a stopping time then for all $t \geq 0$, we have that $(\tau < t) \in \mathcal{F}_t$.
9. If for all $t \geq 0$, $(\tau < t) \in \mathcal{F}_t$ and if the filtration satisfies the condition
$$\mathcal{F}_t = \cap_{\epsilon\geq 0}\mathcal{F}_{t+\epsilon}, \quad t \geq 0$$
then τ is a stopping time with respect to the given filtration.
10. Prove that $\tau = \max\{n \in \mathbb{N} \colon X_n \geq 2a\}$ is not a stopping time.
11. Prove that H_A defined example 4.- of 2.13 is a stopping time. Use the fact that
$$\{H_A \leq t\} = \big(\omega \in \Omega : \inf_{\{s\in\mathbb{Q}^+, s\leq t\}} d(X_s(\omega), A) = 0\big)$$
where $\mathbb{Q}^+$ is the set of positive rational numbers.
12. For two stopping times τ_1, τ_2 prove that $\max\{\tau_1, \tau_2\}$ and $I = \min\{\tau_1, \tau_2\}$ are also stopping times.
13. For any sequence of stopping times $(\tau_n)_{n\in\mathbb{N}}$ prove that
$$I := \inf_n \tau_n, \quad S := \sup_n \tau_n$$
are stopping times.

Chapter 3

Martingales

Definition 3.1. *A* **martingale** *defined in* $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$ *is a stochastic process* $\{X_t : t \in J\}$ *such that*

1. $\{X_t : t \in J\}$ *is adapted to the given filtration,*
2. *for all* $t \in J$, $\mathbb{E}(|X_t|) < \infty$,
3. *for all* $s, t \in J$ *such that* $s < t$ *the following condition is fulfilled:*

$$\mathbb{E}(X_t|\mathcal{F}_s) = X_s.$$

Martingales form a very important class of processes, as will be seen in the development of this course. A **submartingale** (resp. **supermartingale**) verifies the first two properties and $\mathbb{E}(X_t|\mathcal{F}_s) \geq X_s$ (resp. $\mathbb{E}(X_t|\mathcal{F}_s) \leq X_s$).

Some properties and examples:

1. The function $s \to E(X_s)$ is constant for a martingale because, for $s < t$ $\mathbb{E}(X_t) = \mathbb{E}(\mathbb{E}(X_t|\mathcal{F}_s)) = \mathbb{E}(X_s)$.
2. In the random walk if $\mathbb{E}(\xi_1) = 0$, then

$$X_n = a + \sum_{i=1}^{n} \xi_i$$

   is a martingale with respect to the canonical filtration $(\mathcal{F}_n)_{n\in\mathbb{N}}$ and is called discrete time martingale. If $\mathbb{E}(\xi_1) \geq 0$ (or $\mathbb{E}(\xi_1) \leq 0$) the corresponding random walk is a submartingale (or supermartingale) with respect to the canonical filtration. See exercise **??**
3. Any process $\{X_t : t \in [0, T]\}$ such that
   (a) $\mathbb{E}(X_t - X_s) = 0, \quad t > s \geq 0,$
   (b) $X_{t+s} - X_s$ and $\mathcal{F}_s$ are independent, for all $s, t \geq 0$

   is a martingale with respect to the canonical filtration $(\mathcal{F}_t)_t$.

   See the exercises in 3.2.

3.1 Some important properties of Martingales

Let $(X_n)_n$ be a discrete time martingale in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{n}})_{\mathrm{n}\in\mathrm{J}}, \mathbb{P})$.

Proposition 3.2 (Optional sampling theorem or Doob stopping theorem)**.** *Let $(X_n)_{n=0}^k$ be a discrete submartingale defined in $(\Omega, \mathcal{F})$ where consider the canonical filtration $(\mathcal{F}_n)_{n=0}^k$. Let $S, T : \Omega \to \mathbb{N}$ two discrete stopping times such that $S \leq T \leq k$, then*

$$\mathbb{E}(X_T) \geq \mathbb{E}(X_S).$$

If X is a martingale, we have an equality.

Observe that for discrete time martingales, X_T is a random variable.

Proof. $\Omega = \bigcup_{j=0}^k (S = j)$ and it is a disjoint union; so we can write X_S as $X_S(\omega) = \sum_{j=0}^k X_j \mathbb{1}_{(S=j)}(\omega)$. We shall only prove a special case since this is the case we will need: $T \equiv k$ and $S \leq k$.

$$X_T - X_S = \sum_{j=0}^{k} (X_k - X_j)\mathbb{1}_{(S=j)} = \sum_{j=0}^{k-1} (X_k - X_j)\mathbb{1}_{(S=j)}$$

and this sum has positive expectation (or is zero in the martingale case), since the set $(S = j) \in \mathcal{F}_j$, we get

$$\mathbb{E}\left[\mathbb{1}_{(S=j)}(X_k - X_j)\right] \geq 0, \;\; j = 0, 1, \ldots, (k-1).$$

■

Proposition 3.3. *Let $(X)_{n=0}^N$, be a discrete positive submartingale in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{n}})_{\mathrm{n}\in\mathrm{J}}, \mathbb{P})$, then for all $\lambda > 0$*

$$\mathbb{P}\left(\sup_n X_n \geq \lambda\right) \leq \frac{1}{\lambda}\,\mathbb{E}\left(X_N \cdot \mathbb{1}_{(\sup_n X_n \geq \lambda)}\right) \leq \frac{1}{\lambda}\,\mathbb{E}(|X_N|).$$

Proof. Observe that here $J = \{0, 1, \ldots, N\}$. Let

$$S = \begin{cases} \inf\{n : X_n \geq \lambda\}, & \text{if } \{n : X_n \geq \lambda\} \neq \emptyset \\ N \text{ if } \{n : X_n \geq \lambda\} = \emptyset \end{cases} \tag{3.1}$$

S is a stopping time, $S \leq N$, so by proposition 3.2

$$\begin{aligned} \mathbb{E}(X_N) &\geq \mathbb{E}(X_S) \\ &= \mathbb{E}\left(X_S \cdot \mathbb{1}_{(\sup_n X_n \geq \lambda)}\right) + \mathbb{E}\left(X_S \cdot \mathbb{1}_{(\sup_n X_n < \lambda)}\right) \\ &\geq \lambda\,\mathbb{P}\left(\sup_n X_n \geq \lambda\right) + \mathbb{E}\left(X_N \cdot \mathbb{1}_{(\sup_n X_n < \lambda)}\right) \end{aligned}$$

The last inequalities are due to the definition of S, i.e. $X_S \geq \lambda$ on the set $(\sup_n X_n \geq \lambda)$ and $X_S = X_N$ on the set $(\sup_n X_n \leq \lambda)$. ■

For a discrete time martingale X, we then have (by Jensen Inequality) a positive submartingale $(|X_n|^p : n \in \{0, 1, 2, \ldots, N\})$ whenever $\mathbb{E}(|X_N|^p) < \infty$ and in any case we obtain:

Corollary 3.4. *If X is a discrete time martingale indexed by the finite set $\{0, 1, 2, \ldots, N\}$ and if $p \geq 1$ then*

$$\lambda^p \, \mathbb{P}\left[\sup_n |X_n| \geq \lambda\right] \leq \mathbb{E}(|X_N|^p), \;\; \lambda > 0$$

The former properties can be extended to the continuous time case:

Theorem 3.5 (Doob's maximal inequality)**.** *Let $(M_t)_{t\geq 0}$ be a right continuous martingale and $p \geq 1$ then*

$$\mathbb{P}\left(\sup_{0\leq s\leq t} |M_s| \geq \lambda\right) \leq \frac{1}{\lambda^p}\,\mathbb{E}(|M_t|^p), \qquad t \geq 0, \;\; \lambda > 0.$$

Proof. Let D be a countable dense subset of the interval $[0, t]$. Because of the right continuity

$$\sup_{s\in D} |M_s| = \sup_{s\in[0,t]} |M_s|.$$

Let $D_n \subset D$ be an increasing sequence, such that D_n is finite and $D = \bigcup_n D_n$. Since the result holds for each discrete submartingale indexed by a finite set D_n, $(|M_s|)_{s\in D_n}$, we have for all $n \in \mathbb{N}$

$$\mathbb{P}\left[\sup_{s\in D_n} |M_s| \geq \lambda\right] \leq \frac{1}{\lambda^p}\,\mathbb{E}(|M_t|^P).$$

Passing to the limit as $n \to \infty$ and using the dominated convergence theorem,

$$\lim_{n\to\infty} \mathbb{P}\left[\sup_{s\in D_n} |M_s| \geq \lambda\right] = \mathbb{P}\left[\lim_{n\to\infty} (\sup_{s\in D_n} |M_s| \geq \lambda)\right] \leq \frac{1}{\lambda^p}\,\mathbb{E}(|M_t|^P).$$

Finally, since $\lim_{n\to\infty}(\sup_{s\in D_n} |M_s|) = \sup_{s\in D} |M_s|$ we conclude. ■

3.2 Exercises

1. Random Walk (RW) : $S_n := a + \sum_{i=1}^n \xi_i, \;\; a \in \mathbb{R}$ be a RW . Then
   - ♢ If $\mathbb{E}(\xi_1) = 0$ the R.W. is a discrete time martingale.
   - ♢ If $\mathbb{E}(\xi_1) \geq 0$ the R.W. is a discrete time submartingale.
   - ♢ If $\mathbb{E}(\xi_1) \leq 0$ the R.W. is a discrete time super martingale.
2. If $(X_n)_{n\in\mathbb{N}}$ is an adapted process with independent increments, such that $\mathbb{E}(X_n - X_{n-1}) = 0$ for all $n \in \mathbb{N}$, is it true that $(X_n)_{n\geq 0}$ is a martingale? If $(X_t)_{t\geq 0}$ is an adapted process with independent increments, such that $\mathbb{E}(X_{t+h} - X_t) = 0$ for all $h, t \in \mathbb{R}^+$, is it true that $(X_n)_{n\geq 0}$ is a martingale?
3. Prove the Optional sampling theorem with the same notations and hypotheses as in the text, but for any stopping time T.

Chapter 4

Brownian Motion

4.1 Gaussian Random Variables

4.1.1 Some known results on gaussian vectors

We start this section by recalling some basic resultas on Gaussian random variables and Gaussian random processes.

Definition 4.1. *The normal or Gaussian random variable $X : \Omega \to \mathbb{R}$ with mean m and variance σ^2 ($X \approx \mathcal{N}(m, \sigma^2)$), has density*

$$\frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp\left[-\frac{(x-m)^2}{2\sigma^2}\right]$$

Some useful properties on Gaussian or normal random variables are:

1. if $m = 0$, $\mathbb{E}(X^2) = \sigma^2$, $\mathbb{E}(X^4) = 3\sigma^4$,
2. $\mathrm{var}(X^2) = \mathbb{E}\left[X^2 - \sigma^2\right]^2 = 2\sigma^4$,
3. (Laplace transform). $\psi(\lambda) = \mathbb{E}\left[\exp(\lambda X)\right] = \exp(\lambda m + \frac{\sigma^2\lambda^2}{2})$,
4. (Characteristic function). $\Phi_X(u) = \mathbb{E}\left[\exp(iuX)\right] = \exp(imu - \frac{\sigma^2 u^2}{2})$.

Definition 4.2. *The normal or Gaussian random vector $X : \Omega \to \mathbb{R}^n$ is multivariate normal random variable i.e. it verifies the property that any linear combination of its components is also Gaussian.*

Recall that for a Gaussian random vector $(X_k)_{k=1,2,\ldots,n}$ we have its mean vector

$$\{m(i) = \mathbb{E}(X_i)\}_{i=1,2,\ldots,n}$$

and its covariance $n \times n$ matrix $\mathbf{D}$ given by

$$(D(i,j) = \mathbb{E}[(X_i - m(i))(X_j - m(j))])_{i,j\in\{1,2,\ldots,k\}}.$$

D verifies

1. symmetric, i.e. $D(i,j) = D(j,i)$,

2. positive definite, i.e., for any $a_i \in \mathbb{R}$, $i = 1, 2, \ldots, n$,

$$\sum_{1 \le i,j \le n} a_i\, a_j D(i,j) \ge 0.$$

The first property is immediate and the second can easily be proved :

$$\begin{aligned}\sum_{1 \le i,j \le n} a_i\, a_j D(i,j) &= \sum_{1 \le i,j \le n} a_i\, a_j\, \mathbb{E}[(X_i - m(i))\,(X_j - m(j))] \\ &= \mathbb{E}(\sum_{1 \le i,j \le n} a_i\, a_j (X_i X_j)) - \sum_{1 \le i,j \le n} a_i\, a_j m(i) m(j) \\ &= \mathbb{E}[\sum_{1 \le i \le n} a_i X_i]^2 - (\sum_{1 \le i \le n} a_i m(i))^2\end{aligned}$$

And this is positive since it is the variance of the random variable $Y = \sum_{1 \le i \le n} a_i X_i$.

Another well known fact is that given an n-dimensional vector $m = (m_1, \ldots, m_n)$ and a symmetric, positive definite matrix $\mathbf{D} = (D_{i,j})_{i,j \in \{1,2,\ldots,n\}}$, there exists an n dimensional Gaussian vector with mean vector m and covariance matrix $\mathbf{D}$.

Proposition 4.3. *Given a function $m\colon \{1, 2, \cdots n\} \to \mathbb{R}$ and a symmetric positive definite function $D\colon \{1, 2, \cdots n\} \times \{1, 2, \cdots n\} \to \mathbb{R}$ there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and Gaussian vector $(X_1, X_2, \cdots X_n)$ defined in $(\Omega, \mathcal{F}, \mathbb{P})$ with mean function m and covariance function D.*

4.1.2 Gaussian processes

Definition 4.4. *A Gaussian process is a stochastic process $\{X_t : t \in [0, T]\}$ such that, for all $n \in \mathbb{N}$, $\lambda_i \in \mathbb{R} : i \in \{1, 2, \ldots, n\}$ and any $\{t_i \in J,\ i = 1, 2, \ldots, n\}$ the linear combination*

$$\sum_{i=1}^{n} \lambda_i X_{t_i}$$

is a Gaussian random variable.

This is equivalent to say that all the finite dimensional distributions are Gaussian random vectors.

A Gaussian process $(X_t)_{t \in [0,T]}$ can also be characterized by its mean function $m\colon [0, T] \to \mathbb{R}$

$$m(t) = \mathbb{E}(X_t), \qquad t \in [0, T]$$

and its covariance function $\mathbf{C}\colon [0, T] \times [0, T] \to \mathbb{R}$,

$$C(s, t) = \mathbb{E}[(X_s - m(s))(X_t - m(t))]_{s,t \in [0,T]}.$$

The precise result is:

Proposition 4.5. *Given a function $m\colon [0, T] \to \mathbb{R}$ and a symmetric positive definite function $C\colon [0, T] \times [0, T] \to \mathbb{R}$ there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and Gaussian process $(X_t)_{t \in [0,T]}$ defined on it such that it has mean function m and covariance function C.*

This result is based on Daniel-Kolmogorov existence theorem and on the property 4.3 of Gaussian vectors mentioned before.

4.2 The Brownian Motion

This mathematical object is related to problems that evolve in time in very chaotic way, so in the beginning the XX century it seemed difficult to make any mathematical model to describe this. It has proven to be one of the most important processes in the theory and in many applications.

- ♢ In 1826-27 a biologist Robert Brown observed the movement of pollen particles placed in water, and saw it was very irregular and erratic. He described it but could not explain why it moved;
- ♢ Bachelier in 1900 described the fluctuations in stock prices and found a mathematical model for them;
- ♢ later Einstein (re)-discovered it in 1905, when studying the movement of a diffusing particle.
- ♢ Smoluchowski found a description of BM as a limit of R.W.
- ♢ finally in 1923 Norbert Wiener gave a rigorous construction of the BM and studied many of its properties.
- ♢ Kolmogorov, P. Lévy, Doob, M. Yor amongst many others developed this subject during the XX-th century.

Definition 4.6. *A real valued stochastic process $B = (B_t)_{t\geq 0}$ defined in $(\Omega, \mathcal{F}, \mathbb{P})$ is called the standard Brownian motion(BM) or Wiener process if*

1. $B_0 = 0$ *a.s.*
2. $B_{t+h} - B_t$ *has* $\mathcal{N}(0,h)$ *distribution, for* $0 \leq h$ *and* $0 \leq t$.
3. *For all* $n \in \mathbb{N}$ *and all times* $0 \leq t_1 < t_2 < \cdots < t_n$ *the random variables* $(B_{t_i} - B_{t_{i-1}})_{(i=1,2,\ldots,n)}$ *are independent.*

The second property means that the process has **stationary increments** i.e. $B_{t+h} - B_t$ has the same distribution as $B_{s+h} - B_s$ for any $h, s, t \in [0, \infty)$ and $(B_{t+h} - B_t) \approx \mathcal{N}(0,h)$. The third condition can also be stated as:
for all $t, s > 0$, the random variable $B_{t+s} - B_t$ is independent of $\mathcal{F}_s = \sigma\{B_u : u \leq s\}$.

It is an easy exercise to prove that both formulations for the independence of the increments are equivalent, but we will usually only use the first one.

We now state without proof another fundamental theorem of Kolmogorov:

Theorem 4.7. *If we have a process $(X_t)_{t\geq 0}$ defined in some space $(\Omega, \mathcal{F}, \mathbb{P})$, such that for all $T > 0$ there exist positive constants α, β, C with*

$$\mathbb{E}[|X_s - X_t|^\alpha] \leq C \cdot |s-t|^{1+\beta} \quad s, t \in [0,T]$$

then the process admits a continuous version on $[0,T]$.

As a consequence the BM has a continuous version, since the BM verifies the condition

$$\mathbb{E}_x[|B_t - B_s|^4] \leq C(t-s)^2$$

This is immediate from property 2 of.....

Now we will see that BM as a special case of Gaussian processes:

Proposition 4.8. *A standard Brownian motion*$B = (B_t)_{t\geq 0}$ *defined in* $(\Omega, \mathcal{F}, \mathbb{P})$ *verifies the conditions*

(a) It is a Gaussian Process .

(b) $m(t) = \mathbb{E}(B_t) = 0, \quad t \geq 0,$

(c) $cov(B_t, B_s) = C(s,t) = \min(s,t), \quad s,t \geq 0$

and the reciprocal result is also true: a Gaussian process verifying the three former conditions is a BM.

Proof. If B verifies definition (4.6), then,

(a) It is a Gaussian process because, given $n \in \mathbb{N}$, $0 \leq t_1 \leq t_2 \leq \cdots \leq t_n$ for any $a_i \in \mathbb{R}, i = 1, 2, \ldots, n$ there exist $b_i \in \mathbb{R}, i = 1, 2, \ldots, n$ verifying

$$\sum_{i=1}^{n} a_i B_{t_i} = \sum_{i=1}^{n} b_i (B_{t_i} - B_{t_{i-1}}).$$

and since the last expression is the sum of independent gaussian. random variables, the sum is also gaussian.

(b) $m(t) = 0, t \geq 0$ and

(c) for $0 \leq s < t$

$$\operatorname{cov}(B_s, B_t) = \mathbb{E}(B_s B_t) = \mathbb{E}(B_s(B_t - B_s) + B_s^2) = s$$

since $\mathbb{E}(B_s)\,\mathbb{E}(B_t - B_s) = 0$ and $\mathbb{E}(B_s^2) = s$.

Conversely, if is a gaussian process with the stated conditions,

1. $B_0 = 0$ a.s because $\mathbb{E}(B_0^2) = 0$,
2. for $0 \leq s \leq t$, $(B_t - B_s)$ is Gaussian and

$$\mathbb{E}((B_t - B_s)^2) = \mathbb{E}(B_t^2 - 2B_t B_s + B_s^2)) = t - 2\min(s,t) + s = t - s.$$

   So $(B_t - B_s) \approx \mathcal{N}(0, t-s)$

3. to see the independence of the increments, it is enough to prove that the covariance matrix of the Gaussian vector $(B_{t_i} - B_{t_{i-1}})_{i=1}^n$ is diagonal. This is a consequence of

$$\mathbb{E}\left((B_{t_i} - B_{t_{i-1}})(B_{t_j} - B_{t_{j-1}})\right) = t_i - t_i - t_{i-1} + t_{i-1} = 0.$$

   for $t_i \leq t_{j-1} < t_j$.

■

The existence of the BM is a consequence of the proposition for the existence of Gaussian processes. This is done taking the mean function as $m(t) = 0, t \geq 0$ and the covariance function as $C(s,t) = \mathrm{cov}(B_s, B_t) = \min(s,t), \;\; s,t \geq 0$.

All we must verify is that this function is positive definite (symmetry is immediate). So for any $n \in \mathbb{N}$, $s_i \in \mathbb{R}^+$ and $a_i \in \mathbb{R}, \;\; i = 1, 2, \ldots, n$,

$$\begin{aligned}\sum_{1\leq i,j\leq n} a_i a_j C(s_i, s_j) &= \sum_{1\leq i,j\leq n} a_i a_j \int_0^\infty \mathbb{1}_{[0,s_i)}(s)\mathbb{1}_{[0,s_j)}(s)ds \\ &= \int_0^\infty \big(\sum_{i=1}^n a_i \mathbb{1}_{[0,s_i)}(s)\big)^2 ds \geq 0\end{aligned}$$

Definition 4.9. *A process B with values in $\mathbb{R}^n$, is an n-dimensional Brownian motion if it can be written as*

$$B_t = (B_t^{(1)}, B_t^{(2)}, \ldots, B_t^{(n)}), \;\; t \geq 0$$

where $(B_t^{(i)})_{i=1,2,\ldots,n}$ are independent real valued BM. It will also have a continuous version.

Examples 4.10 (Transformed processes of standard BM)**.** *If B is a standard BM then following processes are also standard BM:*

(a) Symmetry: $(Z_t = -B_t)_{t\geq 0}$,

(b) Scaling: for any $c > 0$ the process $(B_t^c = \frac{1}{c}B_{c^2 t} : t \geq 0)$,

(c) Translation by $h > 0$: the process $(Y_t = B_{t+h} - B_h)_{t\geq 0}$,

(d) reversion in time: for $T > 0$, the process $(R_t^T = B_T - B_{T-t})_{t\in[0,T]}$ is also a Brownian motion in $[0,T]$,

(e)

$$Y_t := \begin{cases} tB_{\frac{1}{t}} & \text{if } t > 0 \\ 0 & \text{if } t = 0. \end{cases}$$

Proof. Exercise. ■

Examples 4.11. *The process associated to the Brownian motion $B = (B_t : t \geq 0)$ are martingales with respect to canonical filtration of the BM.*

1. $(B_t : t \geq 0)$

2. $(B_t^2 - t : t \geq 0)$

3. For any $a \in \mathbb{R}$, $\Big(M_t = \exp(aB_t - \frac{a^2}{2}t) : t \geq 0\Big)$.

Proof. Let $0 \leq s < t$.

1. $\mathbb{E}(B_t - B_s|\mathcal{F}_s) = \mathbb{E}(B_t - B_s) = 0$

2. $\mathbb{E}(B_t^2 - B_s^2 - (t-s)|\mathcal{F}_s) = \mathbb{E}\left((B_t - B_s)^2 - (t-s) + 2B_s(B_t - B_s)|\mathcal{F}_s\right) = 0$

3.
$$\mathbb{E}\left(M_t|\mathcal{F}_s\right) = M_s \text{ iff } \mathbb{E}\left(\frac{M_t}{M_s}|\mathcal{F}_s\right) = 1$$

and this last conditional expectation equals (using the properties of Gaussian r.v)

$$\begin{aligned}\mathbb{E}\left[\exp(a(B_t - B_s)) - \frac{a^2}{2}(t-s))|\mathcal{F}_s\right] &= \exp(-\frac{a^2}{2}(t-s))\,\mathbb{E}\left[\exp(a(B_t - B_s)\right] \\ &= 1\end{aligned}$$

■

Definition 4.12. *The BM* **started at point** $y \in \mathbb{R}$. *It is the translated process* $(y+B_t : t \geq 0)$. *We use the notations* $\mathbb{E}_y(\cdot)$, $\mathbb{P}_y(\cdot)$ *for the expectation and the probability in this case. For* $A \in \mathcal{B}(\mathbb{R})$ *the definition of BM gives us the distribution of each* B_t,

$$\mathbb{P}(B_t \in A) = \int_A p(x,0,t)dx = \frac{1}{(2\pi t)^{\frac{1}{2}}}\int_A \exp\left[-\frac{x^2}{2t}\right]dx,$$

$$\mathbb{P}_y(B_t \in A) = \int_A p(x,y,t)dx = \frac{1}{(2\pi t)^{\frac{1}{2}}}\int_A \exp\left[-\frac{(x-y)^2}{2t}\right]dx,$$

and for any borel-measurable and bounded function $f : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}_y(f(B_t)) = \int f(x)p(x,y,t)dx = \frac{1}{(2\pi t)^{\frac{1}{2}}}\int f(x)\exp\left[-\frac{(x-y)^2}{2t}\right]dx,$$

4.2.1 Computation of joint probabilities.

Theorem 4.13. *For* $n \in \mathbb{N}$, $0 \leq t_1 \leq t_2 \leq \cdots \leq t_n$ *and* $f : \mathbb{R}^n \to \mathbb{R}$, *bounded and measurable,*

$$\begin{aligned}\mathbb{E}[f(B_{t_1}, B_{t_2}, \ldots, B_{t_n})] = \int_{-\infty}^{+\infty}\cdots\int_{-\infty}^{+\infty} & f(x_1, x_2, \ldots, x_n)\, p(x_1, 0, t_1)\, p(x_2, x_1, t_2 - t_1)\cdots \\ & \cdots p(x_n, x_{n-1}, t_n - t_{n-1})\, dx_n \cdots dx_2 dx_1\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}_x[f(B_{t_1}, B_{t_2}, \ldots, B_{t_n})] = \int_{-\infty}^{+\infty}\cdots\int_{-\infty}^{+\infty} & f(x_1, x_2, \ldots, x_n)\, p(x_1, x, t_1)\, p(x_2, x_1, t_2 - t_1)\cdots \\ & \cdots p(x_n, x_{n-1}, t_n - t_{n-1}) dx_n \cdots dx_2 dx_1\end{aligned}$$

Idea of the proof. There exists a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ such that

$$h(y_1, \ldots, y_n) := f(T(y_1, \ldots, y_n)) = f(y_1, y_1 + y_2, \ldots, y_1 + y_2, \cdots + y_n)$$

then define $Y_i = B_{t_i} - B_{t_{i-1}}, i = 1, 2, \ldots, n$ which are independent, so they have a joint density, $p(y_1, 0, t_1)p(y_2, 0, t_2 - t_1)\cdots p(y_n, 0, t_n - t_{n-1})$ and

$$\begin{aligned}\mathbb{E}[f(B_{t_1}, B_{t_2}, \ldots, B_{t_n})] = \mathbb{E}[h(Y_{t_1}, Y_{t_2}, \ldots, Y_{t_n})] \int_{-\infty}^{+\infty}\cdots\int_{-\infty}^{+\infty} & h(y_1, y_2, \ldots y_n)\cdots \\ \cdots p(y_1, 0, t_1)\, p(y_2, 0, t_2 - t_1)\cdots p(y_n, 0, t_n - t_{n-1}) dy_n \cdots dy_2\, dy_1 & \end{aligned}$$

Since the Jacobian of T equals 1, we can transform this integral in order to obtain the result. ■

4.2.2 Markov Property

Theorem 4.14. *Let B be a standard Brownian motion. For a borel and bounded function, $f\colon \mathbb{R} \to \mathbb{R}$ and $0 \le s \le t$*

1. $\mathbb{E}(f(B_t)|\mathcal{F}_s) = \mathbb{E}(f(B_t)|B_s)$,
2. $\mathbb{E}\left[f(B_t)|\sigma(B_s)\right] = \dfrac{1}{(2\pi(t-s))^{\frac{1}{2}}} \displaystyle\int_{\mathbb{R}} f(x) \exp\left[-\frac{(x-B_s)^2}{2(t-s)}\right] dx$,
3. $\mathbb{E}(f(B_t)|B_s = y) = \dfrac{1}{(2\pi(t-s))^{\frac{1}{2}}} \displaystyle\int_{\mathbb{R}} f(x) \exp\left[-\frac{(x-y)^2}{2(t-s)}\right] dx$,

Proof. We will prove the second equality of the theorem: We call

$$\Psi(y) := \int_{\mathbb{R}} f(x) p(x, y, t-s)\, dx.$$

To prove the result we show that for all $C \in \sigma(B_s)$,

$$\mathbb{E}(f(B_t)\mathbb{1}_C) = \mathbb{E}(\Psi(B_s)\mathbb{1}_C),$$

i.e. $\mathbb{E}(f(B_t)|\sigma(B_s)) = \Psi(B_s)$. But C is of the form $(B_s \in A)$ for some $A \in \mathcal{B}(\mathbb{R})$. Hence, the left hand side (by theorem 3) is equal to

$$\begin{aligned}
\mathbb{E}\left[f(B_t)\mathbb{1}_{(B_s \in A)}\right] &= \int_{\mathbb{R}} \int_{\mathbb{R}} f(x)\mathbb{1}_A(y) p(y, 0, s) p(y, x, t-s)\, dx\, dy \\
&= \int_A p(y, 0, s)\Psi(y)\, dy
\end{aligned}$$

The right hand side

$$\mathbb{E}(\Psi(B_s)\mathbb{1}_C) = \int_\Omega \mathbb{1}_A(B_s) \cdot \Psi(B_s) d\mathbb{P} = \int_A p(y, 0, s)\Psi(y)\, dy$$

(the last equality is just the change of variable formula). To prove (1) we define $F(x, y) := f(x+y),\ \ x, y \in \mathbb{R}$

$$\begin{aligned}
\mathbb{E}\left(f(B_t)|\mathcal{F}_s\right) &= \mathbb{E}\left(f(B_t - B_s + B_s)|\mathcal{F}_s\right) \\
&= \mathbb{E}\left(F(B_t - B_s, B_s)|\mathcal{F}_s\right) \\
&= \frac{1}{(2\pi(t-s))} \int_{\mathbb{R}} F(x, B_s) \exp\left[-\frac{x^2}{2(t-s)}\right] dx
\end{aligned}$$

We now define $\Phi(y) := \dfrac{1}{(2\pi(t-s))^{\frac{1}{2}}} \displaystyle\int_{\mathbb{R}} F(x, y) \exp\left[-\frac{x^2}{2(t-s)}\right] dx$ and obtain,

$$\begin{aligned}
\Phi(y) &= \frac{1}{(2\pi(t-s))^{\frac{1}{2}}} \int_{\mathbb{R}} f(x+y) \exp\left[-\frac{x^2}{2(t-s)}\right] dx \\
&= \frac{1}{(2\pi(t-s))^{\frac{1}{2}}} \int_{\mathbb{R}} f(x) \exp\left[-\frac{(x-y)^2}{2(t-s)}\right] dx
\end{aligned}$$

So

$$\mathbb{E}\left(f(B_t)|\mathcal{F}_s\right) = \Phi(B_s) = E\left[f(B_t)|B_s\right]$$

■

4.3 Exercises

1. Prove that processes given in example (4.10) are Brownian Motions.

2. Let B_t a standard Brownian motion.

   a) Use the fact that B_t is a Gaussian process to prove

   $$\mathbb{E}\left[e^{iuB_t}\right] = \exp\left(-\frac{1}{2}u^2 t\right)$$

   b) Use the power expansion of the exponential function on both sides, compare the terms with the same power of u and deduce that

   $$\mathbb{E}\left[B_t^4\right] = 3t^2$$

   and more generally that

   $$\mathbb{E}\left[B_t^{2k}\right] = \frac{(2k)!}{2^k \cdot k!} t^k \ ; \quad t \in \mathbb{N}$$

   c) To prove b) with another more rigorous method you can proceed as follows: Use the distribution of B_t to calculate $\mathbb{E}[f(B_t)]$, apply this to $f(x) = x^{2k}$ and use integration by parts and induction on k.

   d) Use b) and theorem (4.7) to prove that B_t admits a continuous version.

3. Given a one dimensional Brownian Motion $(B_t)_t$. Then,

   a) Prove that Brownian bridge $Z_t := (B_t - tB_1)$, $t \in [0,1]$ is a Gaussian process, compute its mean and variance.

   b) If $U_t := \exp(-tB_{2t})$, compute $\mathbb{E}[U_t \cdot U_s]$

4. Given a n-dimensional Brownian Motion $\left[\mathbf{B}_t = \left(B_t^{(1)}, B_t^{(2)}, \ldots, B_t^{(n)}\right)\right]_{t>0}$, prove that for $i, j = 1, 2, \ldots, n$,

   $$\mathbb{E}[B_t^{(i)} \cdot B_s^{(j)}] = \min(s,t)\delta_{i,j}$$

   and

   $$\mathbb{E}\left[\left(B_t^{(i)} - B_s^{(i)}\right)\left(B_t^{(j)} - B_s^{(j)}\right)\right] = (t-s)\delta_{i,j} \ ; \quad t > s > 0$$

Chapter 5

Stochastic Itô Integral

5.1 Discrete case: Integration w.r.t. a Simple symmetric Random Walk

Let $(\xi_n)_{n\in\mathbb{N}}$ be a sequence of independent Bernoulli random variables, i.e. $\mathbb{P}(\xi_1 = 1) = \frac{1}{2} = \mathbb{P}(\xi_1 = -1)$, then the corresponding **random walk**

$$X_n = \sum_{i=1}^{n} \xi_i, \qquad \Delta X_i = X_i - X_{i-1} = \xi_i$$

We can consider each ξ_i as the result of a game and we can try to find a good strategy for the game by betting each time a certain amount A_i but we have to make the decision before the next play, with just the information of what has happened until time $(i-1)$ i.e. A_i must be $\sigma(X_k : k \leq i-1)$-measurable for each $i \in \mathbb{N}$, and the capital up to time n, will be

$$Z_n = \sum_{i=1}^{n} A_i \xi_i = \sum_{i=1}^{n} A_i (X_i - X_{i-1}) = \sum_{i=1}^{n} A_i \Delta X_i$$

It is easy to see that

Proposition 5.1.

1. *$(Z_n : n \in \mathbb{N})$ is a martingale for the canonical filtration $\mathcal{F}_n = \sigma(X_k : k \leq n)$.*
2. *$\mathbb{E}(Z_n) = 0$, $\operatorname{var}(Z_n) = \sum_{i=1}^{n} \mathbb{E}(A_i^2)$ if $A_n \in L_2(\Omega, \mathcal{F}, \mathbb{P})$.*

Proof. It is a martingale: for any $n \in \mathbb{N}$,

$$\mathbb{E}(Z_n | \mathcal{F}_{n-1}) = \mathbb{E}\left[(Z_{n-1} + A_n \xi_n) | \mathcal{F}_{n-1}\right] = Z_{n-1},$$

because $\mathbb{E}(A_n \xi_n | \mathcal{F}_{n-1}) = A_n \mathbb{E}(\xi_n) = 0$ since A_n is $\mathcal{F}_{n-1}$-measurable and ξ_n is independent of $\mathcal{F}_{n-1}$ and has mean 0. To compute its variance:

$$\mathbb{E}\left(Z_n^2\right) = \mathbb{E}\left(\sum_{i,j}^{n} A_i A_j \xi_i \xi_j\right) = \mathbb{E}\left(\sum_{i}^{n} A_i^2 \xi_i^2\right) = \sum_{i}^{n} \mathbb{E}\left(A_i^2\right) \mathbb{E}\left(\xi_i^2\right) = \sum_{i}^{n} \mathbb{E}\left(A_i^2\right),$$

we used:

◇ for the second equality, if $i < j$,

$$\begin{aligned}\mathbb{E}(A_iA_j\xi_i\xi_j) &= \mathbb{E}(\mathbb{E}(A_iA_j\xi_i\xi_j|\mathcal{F}_{j-1}))\\ &= \mathbb{E}(A_iA_j\xi_i\,\mathbb{E}(\xi_j|\mathcal{F}_{j-1}))\\ &= \mathbb{E}(A_iA_j\xi_i\,\mathbb{E}(\xi_j))\\ &= 0,\end{aligned}$$

because of the measurability and independence assumed.

◇ for the third one, A_i depends only on $(\xi_k : k = 1, 2, \ldots, i-1)$ and these are independent of ξ_i.

◇ $\mathbb{E}(\xi_i^2) = 1$.

■

More generally, given a discrete Martingale $(X_n)_{n\in\mathbb{N}}$, in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{n}})_{\mathrm{n}\in\mathrm{J}}, \mathbb{P})$ and a bounded positive process $A = (A_n)_{n\in\mathbb{N}}$ such that A_n is $\mathcal{F}_{n-1}$-measurable, $n = 1, 2, \ldots$ we can define the stochastic integral (also called the martingale transform of A) as:

$$(A \cdot X)_n = Z_n = \sum_{i=1}^{n} A_i(X_i - X_{i-1}) = \sum_{i=1}^{n} A_i\Delta X_i, \qquad n \in \mathbb{N}\,.$$

It can be proved that it is also a martingale and we can compute its mean and variance (exercise).

In the continuous case, we want to give sense to

1. $Z_t = \displaystyle\int_0^t g(s)dB_s$, for $g\colon \mathbb{R} \to \mathbb{R}$
2. to $\displaystyle\int h(X_s)dB_s$ for a process $(X_s : s \geq 0)$,
3. or to $\displaystyle\int H(s, X_s)dB_s$ for a process $(X_s : s \geq 0)$.

for convenient classes of functions g, h, H and processes X. The first one is rather easy to study: **the Paley-Wiener** integral, and is defined as

$$\int_0^t g(s)dB_s = -\int_0^T g'(t)B_t dt$$

, for g differentiable and such that $g(0) = g(T) = 0$. It can be proved for $g \in \mathcal{C}^{(1)}$ and using integration by parts that we also have,

$$\int_0^t g(s)dB_s = g(T)B_T - \int_0^T g'(t)B_t dt.$$

The second one, will prove to be quite tricky due to the great irregularity of the paths of the Brownian motion $t \to B_t$, as will be seen in the next sections.

This concept is needed in order to develop a theory of stochastic differential equations (SDE) of the form:

$$\begin{aligned} dX_t &= a(t, X_t)dt + b(t, X_t)dB_t \\ X_0 &= x \end{aligned}$$

which must be understood as

$$X_t = x + \int_0^t a(s, X_s)ds + \int_0^t b(s, X_s)dB_s.$$

So our task is to give a meaning to the last integral.

5.2 Quadratic variation of Brownian motion.

Definition 5.2. *Let $(X_t : t \geq 0)$ be a continuous stochastic process. We define its* **p-variation** *on $[a, b] \subset [0, \infty)$ as the limit in probability when $|\pi_n| \to 0$ (if it exist) of*

$$Q_n[a, b] := \sum_{\pi_n} |X_{t^n_{k+1}} - X_{t^n_k}|^p = \sum_{\pi_n} |\Delta X(t^n_k)|^p$$

where $\{\pi_n : n \in \mathbb{N}\}$ is a family of partitions of $[a, b]$, of the form $\pi_n := \{a = t^n_0 < t^n_1 < \cdots < t^n_{m_n} = b\}$ such that $\lim_{n\to\infty} |\pi_n| = 0$.

Remark 5.3. *Recall that $|\pi_n| = \max_k |t^n_{k+1} - t^n_k|$.*

When $p = 2$ it is called **quadratic variation** of X and for $p = 1$ the **total variation** of X in $[a, b]$;

Notation. $\Delta X(t^n_k) := X_{t^n_{k+1}} - X_{t^n_k}$.

We recall the fact: (convergence in $L_2(\Omega, \mathcal{F}, \mathbb{P})$) implies (convergence in probability.)

Proposition 5.4. *The quadratic variation on $[0, t]$ of the Brownian motion, B is equal to t and is denoted $< B, B >_t$.*

Proof. Let $\{\pi_n : n \in \mathbb{N}\}$ be a family of partitions of $[a, b]$, of the form

$$\pi_n := \{a = t^n_0 < t^n_1 < \cdots < t^n_{m_n} = b\},$$

such that $\lim_{n\to\infty} |\pi_n| = 0$. We will use the result: If $X \approx \mathcal{N}(0, \sigma^2)$ then $\mathrm{var}(X^2) = 2\sigma^4$. So if $\Delta B(t^n_k) := B_{t^n_{k+1}} - B_{t^n_k}$, $\quad \Delta t^n_k := (t^n_{k+1} - t^n_k)$ then $\Delta B_{t^n_k} \approx \mathcal{N}(0, \Delta t^n_k)$ and $\mathrm{var}[(\Delta B_{t^n_k})^2] = 2(\Delta t^n_k)^2$. We will prove that

$$L_2 - \lim_{n\to\infty} Q_n = L_2 - \lim_{n\to\infty} \sum_{\pi_n} \left(\Delta B_{t^n_k}\right)^2 = (b - a).$$

$$\mathbb{E}[(Q_n - (b - a))^2] = \mathbb{E}\left[\left(\sum_{\pi_n} \Delta B^2_{t^n_k} - \Delta t^n_k\right)^2\right]$$

This is the variance of a sum of independent random variables, so it is equal to the sum of the variances:

$$= \sum_{\pi_n} \operatorname{var}\left[\Delta B_{t_k^n}^2 - \Delta t_k^n\right] = 2\sum_{\pi_n} (\Delta t_k^n)^2 \leq 2|\pi^n| \cdot (b-a),$$

and since $|\pi_n| \to 0$ as $n \to \infty$, the former goes to 0. ■

The case stated in the proposition is for $a = 0, b = t$.

Corollary 5.5.

1. *If $1 \leq p < 2$ the p-variation of the BM is a.s. infinite.*
2. *The 1-variation of the paths of BM is a.s. infinite.*

Proof. We will do the case $(a, b) = (0, t)$ and we suppose that the BM has finite p-variation for some $p \in [1, 2)$. Let $\delta = 2 - p > 0$. Thanks to our supposition, there exists a subsequence of partitions π_{n_k} for which the convergence of $\left[\sum_{\pi_{n_k}} |B_{t_{k+1}^n}(\omega) - B_{t_k^n}(\omega)|^p\right]$ to a finite number, holds true a.s when $\lim_k |\pi_{n_k}| = 0$. Since

$$\sum_{\pi_{n_k}} |\Delta B_{t_k^n}|^2$$

still converges in $L_2(\Omega, \mathcal{F}, \mathbb{P})$, then we can further obtain a subsequence which we call π'_{n_k} for which both sums converge a.s. So there exists a subset Ω_0 of Ω such that $\mathbb{P}(\Omega_0) = 1$ and for each $\omega \in \Omega_0$,

$$\begin{aligned}\sum_{\pi'_{n_k}} |B_{t_{k+1}^n}(\omega) - B_{t_k^n}(\omega)|^2 &= \sum_{\pi'_{n_k}} |B_{t_{k+1}^n}(\omega) - B_{t_k^n}(\omega)|^p |B_{t_{k+1}^n}(\omega) - B_{t_k^n}(\omega)|^\delta \\ &\leq \sup_{\pi'_{n_k}} |B_{t_{k+1}^n}(\omega) - B_{t_k^n}(\omega)|^\delta \cdot \sum_{\pi_{n_k}} |B_{t_{k+1}^n}(\omega) - B_{t_k^n}(\omega)|^p.\end{aligned}$$

This inequality leads to a contradiction since the BM is continuous. ■

5.3 Stochastic Itô Integral

A simple case: we will try to compute $\int_0^t BdB$ where B is the BM.

As **Riemann sum** it would be

$$\sum_{\pi} B_{r_k}(B_{t_{k+1}} - B_{t_k})$$

where π is a partition: $\{0 = t_0 < t_1 < \cdots < t_m = t\}$ of $[0, t]$ and $r_k \in [t_k, t_{k+1}]$. Since the Brownian motion has paths with unbounded variation, we cannot expect to obtain a pathwise limit.

We will see that different choices of the intermediate point give rise to different limits.

Lemma 5.6. *Let $(\pi_n)_n$ be sequence of a partitions π_n of $[0, t]$ given by $\{0 = t_0^n < t_1^n < \cdots < t_{m_n}^n = t\}$ and such that $\lim_{n\infty} |\pi_n| = 0$.*

Let $\lambda \in [0, 1)$ fixed and define

$$R_n(\lambda) = \sum_{\pi_n} B_{r_k^n} \Delta B_{t_k^n}$$

with $r_k^n = (1-\lambda)t_k^n + \lambda t_{k+1}^n$. *Then*

$$L_2 - \lim_{n\to\infty} R_n(\lambda) = \frac{B_t^2}{2} + (\lambda - \frac{1}{2})t.$$

This means that the limit of the Riemann sum depends on the intermediate point r_k^n selected.

Proof. An easy computation shows that, with the former notations, R_n can be written as $R_n = \dfrac{B_t^2}{2} + A_n + B_n + C_n$ with

1. $A_n = -\dfrac{1}{2}\displaystyle\sum_{\pi_n}(B_{t_{k-1}^n} - B_{t_k^n})^2$
2. $B_n = \displaystyle\sum_{\pi_n}(B_{r_k^n} - B_{t_k^n})^2,$
3. $C_n = \displaystyle\sum_{\pi_n}(B_{t_{k+1}^n} - B_{r_k^n})(B_{r_k^n} - B_{t_k^n})$

It is easy to prove that

- ♢ $L_2 - lim_{n\to\infty} A_n = -\frac{t}{2},$
- ♢ $L_2 - lim_{n\to\infty} B_n = \lambda t,$
- ♢ $L_2 - lim_{n\to\infty} C_n = 0$

The first two properties are an immediate consequence of 5.4 (observe that the partition

$$0 = t_0 \le r_0 = \lambda t_1^n \le \lambda t_1 + (r_1^n - t_1^n) = \lambda t_2^n \le \cdots \le \lambda t_{m_n}^n = \lambda t$$

is a partition of the interval $[0, \lambda t]$). The third one uses the the properties of the independence of the increments.

The case $\lambda = 0$, i.e. $r_k^n = t_k^n$ is easier to prove:

$$R_n(0) = \sum_{\pi_n} B_{t_k^n}\Delta B_{t_k^n} = \frac{1}{2}\sum_{\pi_n}\left[(B_{t_{k+1}^n}^2 - B_{t_k^n}^2) - (B_{t_{k+1}^n} - B_{t_k^n})^2\right]$$

the first term is a telescopic sum and is equal to $\dfrac{B_t^2}{2}$ and we already know that the second converges in L_2 to $-\dfrac{t}{2}$ ■

When we choose r_k^n as the lower extreme of the interval (i.e $\lambda = 0$) we will prove that the limit exists for a large class of processes and it gives rise to the **Itô Stochastic Integral**, which is the one we will develop here. In this case we have computed ,

$$(B\cdot B)_t := \int_0^t BdB = \int_0^t B_s dB_s = \frac{B_t^2}{2} - \frac{t}{2}$$

In the case $\lambda = \frac{1}{2}$, according to the lemma, the Riemann sum converges to

$$R_n\left(\frac{1}{2}\right) \to \frac{B_t^2}{2},$$

This integral is also used and is called the **Stratonovich Integral**.

Observe that: $\left(\frac{B_t^2}{2} - \frac{t}{2},\ t \geq 0\right)$ is a martingale while the Stratonovich integral $\frac{B_t^2}{2}$, is not.

Now we will define the class of processes for which the integral can be defined.

Definition 5.7. *Given* $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in J}, \mathbb{P})$, *and* $0 \leq a < b$ *the class* $\Gamma_2[a,b]$ *is the set of processes* $\{F\colon [0,\infty) \times \Omega \to \mathbb{R}\}$ *such that*

1. F *is progressively measurable,*
2. $\mathbb{E}\left[\int_a^b F(s,\omega)^2 ds\right]$ *is finite.*

Definition 5.8. *A process* $G \in \Gamma_2[0,T]$ *is called elementary process if there exists a fixed partition* $\pi = \{0 = t_0 < t_1 < \cdots < t_m = T\}$ *and random variables* $G_k\colon \Omega \to \mathbb{R}$, $k = 0,1,2,\ldots,m-1$ *such that*

$$G(t,\omega) = G_k(\omega), \ \ \textit{for } t \in [t_k, t_{k+1}), \quad k = 0,1,\ldots,m-1.$$

The set of elementary processes in $[0,T]$ *will be denoted* $\mathcal{E}[0,T]$.

The **Itô stochastic Integral** *of an elementary process* G *(with the former notation)* $[0,T]$ *is defined as*

$$\int_0^T G dB = \sum_{k=0}^{m-1} G_k (B_{t_{k+1}} - B_{t_k}),$$

Remarks 5.9.

- ♢ $\mathbb{E}(G_k^2) < \infty$ *and* G_k *is* $\mathcal{F}_{t_k}$*-measurable,* $k = 1,2,\ldots,m-1$, *since the process is adapted (see the analogy with the betting strategy for RW).*
- ♢ $(G \cdot B)_t$ *or* $< G, B >_t$ *are usual notation for the stochastic integral* $\int_0^t G dB$.
- ♢ *for a subinterval* $[u,v] \subset [0,T]$ *we define* $\int_u^v G dB = \int_0^T G \mathbb{1}_{[u,v)} dB = \left(G \mathbb{1}_{[u,v)} \cdot B\right)_T$.
- ♢ $\int_0^T G dB$ *is also the martingale transform of* $(G_k)_k$ *(seen as a discrete process) w.r.t. the martingale* $(X_k = B_{t_k}, j = 1,2,\ldots,m)$

Now we see the properties of this integral:

Proposition 5.10. *For* $G, G_1, G_2 \in \mathcal{E}[0,T]$, *the stochastic integral*

1. *is additive:* $\int_u^t G dB = \int_u^s G dB + \int_s^t G dB, \ \ 0 \leq u < s < t < T$
2. *is linear:* $\int_0^t (aG_1 + bG_2) dB = a \int_0^t G_1 dB + b \int_0^t G_2 dB, \quad a, b \in \mathbb{R}$
3. *it verifies* $\mathbb{E}\left(\int_0^t G dB\right) = 0$.

4. *the stochastic integral as a process:* $(Z_t := \int_0^t G dB : t \in [0,T])$ *is a continuous square integrable martingale.*

The proof is immediate and is left as an exercise.

Proposition 5.11. *For any elementary process G (with the notations of definition 5.8) and $u, t \in [0,T]$, $u < t$, we have,*

$$\mathbb{E}\left[\left(\int_u^t G dB\right)^2\right] = \mathbb{E}\left[\int_u^t G^2 ds\right]$$

Proof. We do it in the case $u = 0, t = T$ and use the notation $\Delta B_k := B(t_{k+1}) - B(t_k)$.

1. $\mathbb{E}[\Delta B_k] = 0$ and
2. $\mathbb{E}\left[(\Delta B_k)^2\right] = (t_{k+1} - t_k)$,
3. $k \neq j, j < k$.
$$\mathbb{E}[G_k G_j \Delta B_k \Delta B_j] = \mathbb{E}[G_k G_j \Delta B_j]\,\mathbb{E}[\Delta B_k] = 0,$$
4. $k = j$,
$$\mathbb{E}\left[G_k^2 (\Delta B_k)^2\right] = \mathbb{E}\left[G_k^2\right]\mathbb{E}\left[(\Delta B_k)^2\right] = \mathbb{E}(G_k^2)(t_{k+1} - t_k)$$

Now we use these results to compute:

$$\mathbb{E}\left[\left(\int_0^T G dB\right)^2\right] = \sum_{j,k} \mathbb{E}[G_k G_j \Delta B_k \Delta B_j] = \sum_k \mathbb{E}(G_k^2)(t_{k+1} - t_k)$$

which is equal to $\mathbb{E}\left[\int_0^T G^2 ds\right]$. This proves the Itô isometry for elementary processes. For the case $u, t \in [0,T]$ we can do exactly the same with the modified partition $\pi = \{u = t_0 < t_1 < \cdots < t_n = t\}$ or else do it for the elementary processes $G\mathbb{1}_{[u,t)}$. ■

Remark 5.12. *Observe that* $\mathbb{E}\left[\int_O^T G^2 ds\right]$ *is the $L_2([0,T] \times \Omega)$-norm of G.*

The class of elementary processes $\mathcal{E}[0,T]$ is a subset of the product space $L_2(([0,T] \times \Omega), \mathfrak{A}, \mu)$ where $\mathfrak{A}$ is the product σ-algebra $\mathcal{B}([0,T]) \otimes \mathcal{F}$ and μ is the product measure $\lambda \times \mathbb{P}$, with λ the Lebesgue measure in $\mathcal{B}([0,T])$.

Theorem 5.13. *The closure of $\mathcal{E}[0,T]$ with respect to the norm in the product space $L_2([0,T] \times \Omega)$ is $\Gamma_2[0,T]$.*

Proof. The measurability conditions are preserved since an L_2-limit admits a subsequence that is a.s. convergent.

Now, given a process X in $\Gamma_2[0,T]$ the method to find the sequence of elementary processes that converge in the product space to X relies on a deterministic lemma:

Lemma 5.14 (Deterministic lemma). *In the space $L_2[0,T]$ of functions $f : [0,T] \to \mathbb{R}$ we define the operator $P_n : L_2[0,T] \to L_2[0,T]$ by*

$$P_n f(t) = \sum_{j=1}^n \left[\xi_{j-1} \mathbb{1}_{[t_{j-1}, t_j)}\right], \quad \xi_{j-1} = \frac{n}{T} \int_{t_{j-2}}^{t_{j-1}} f(s) ds$$

with $(t_j)_{j=0}^n$ the uniform partition of $[0,T]$: $\left(t_j = \frac{jT}{n}\right)$. Then

1. *for ϕ a step function in $L_2[0,T]$, $\lim_n |P_n\phi - \phi|_2 = 0$,*
2. *the set of step functions is a dense subset of $L_2[0,T]$,*
3. *if $\psi \in L_2[0,T]$ then $\lim_n |P_n\psi - \psi|_2 = 0$.*

Given an element F of $\Gamma_2[0,T]$, (we can suppose that it is bounded by the truncation method). Then for each $\omega \in \Omega$ we construct $P_nF(t,\omega)$, and verify that it is an elementary process. It converges for each ω fixed to $F(\,\cdot\,,\omega)$ in the $L_2[0,T]$-norm, i.e.

$$\lim_n \int_0^T |P_nF(s,\omega) - F(s,\omega)|^2\, ds = 0.$$

Since everything is bounded this is also true when taking expectation:

$$\lim_n \mathbb{E}\left[\int_0^T |P_nF(s,\omega) - F(s,\omega)|^2\, ds\right] = 0. \tag{5.1}$$

We call $G_n = P_nF \in \Gamma_2[0,T]$. Thanks to 5.1 $(G_n)_n$ is a Cauchy sequence in the product space $L_2([0,T]\times\Omega)$ and by the Itô isometry for elementary processes, for any $t \in [0,T]$, the sequence $\left(\int_0^t G_n dB\right)_n$ is a Cauchy sequence in $L_2(\Omega,\mathcal{F},\mathbb{P})$:

$$\begin{aligned}\mathbb{E}\left[\left|\int_0^t G_n dB - \int_0^t G_m dB\right|^2\right] &= \mathbb{E}\left[\left|\int_0^t (G_n - G_m) dB\right|^2\right] \\ &= \mathbb{E}\left[\int_0^t (G_n - G_m)^2 dt\right]\end{aligned}$$

Since $L_2(\Omega,\mathcal{F},\mathbb{P})$ is a complete metric space (in fact a Hilbert space), the $L_2(\Omega,\mathcal{F},\mathbb{P})$ limit of $\left(\int_0^t G_n dB\right)_n$ exists and is by definition **the stochastic integral** of F:

$$(F\cdot B)_t = \int_0^t F dB = L_2 - \lim_{n\to\infty} \int_0^t G_n dB,$$

and the limit does not depend on the approximating sequence. ■

Corollary 5.15. *The Itô Isometry in $\Gamma_2[0,T]$: For any process $F \in \Gamma_2[0,T]$, and $t \in [0,T]$,*

$$\mathbb{E}\left[\left(\int_0^t F dB\right)^2\right] = \mathbb{E}\left[\int_0^t F(s)^2 ds\right]$$

and if $F_n, F \in \Gamma_2[0,T]$ $n \in \mathbb{N}$ are such that

$$\lim_{n\to\infty} \mathbb{E}\left[\int_0^t |F_n - F|^2 dt\right] = 0$$

then $\left(\int_0^t F_n dB\right)_n$ converges in $L_2(\Omega,\mathcal{F},\mathbb{P})$ to $\int_0^t F dB$.

The same properties we had in $\mathcal{E}[0,T]$ hold true in $\Gamma_2[0,T]$:

Theorem 5.16. *If $F, G \in \Gamma_2[0,T]$, $A \in \mathbb{R}$, and $0 \le a < b < c \le T$,*

- ♢ *(additivity)* $\int_a^c F dB = \int_a^b F dB + \int_b^c F dB$ *a.s.*

- ♢ *(linearity)* $\int_a^c (AF+G)dB = A\int_a^c FdB + \int_a^c GdB$
- ♢ $\mathbb{E}\left(\int_a^c FdB\right) = 0$
- ♢ $\int_0^t FdB$ *is* $\mathcal{F}_t$*-measurable, for all* $t \in [0,T]$

The proof is based in the fact that these results are true for elementary processes. In the former result we did not include the last property for elementary processes

Theorem 5.17. *For* $F \in \Gamma_2[0,T]$*, the process defined by*

$$(t,\omega) \to \int_0^t FdB, \quad t \in [0,T]$$

admits a continuous version and it is a square integrable martingale.

Proof. Let (G_n) be the sequence of elementary process that approximate F in $\Gamma_2[0,T]$ and we take any $t \in [0,T]$. Each $\int_0^t G_n dB$ is a continuous martingale, and so $\int_0^t G_n dB - \int_0^t G_m dB$ is also a continuous martingale for any $n, m \in \mathbb{N}$. Thanks to the Doob maximal inequality for martingales (theorem **??**):

$$\begin{aligned}
&\mathbb{P}\left(\sup_{0\le t\le T}\left|\int_0^t G_n dB - \int_0^t G_m dB\right| > \epsilon\right) \\
&\quad \le \frac{1}{\epsilon^2}\,\mathbb{E}\left(\left|\int_0^t G_n dB - \int_0^T G_m dB\right|^2\right) \\
&\quad = \frac{1}{\epsilon^2}\,\mathbb{E}\left(\int_0^T (G_n - G_m)^2 ds\right) \to 0, \quad n,m \to \infty.
\end{aligned}$$

So we can find a subsequence (n_k) such that

$$\mathbb{P}(A_k) := \mathbb{P}\left[\sup_{0\le t\le T}\left|\int_0^t G_{n_{k+1}} dB - \int_0^t G_{n_k} dB\right| > 2^{-k}\right] \le 2^{-k}$$

And by the Borel Cantelli lemma

$$\mathbb{P}(\limsup A_k) = 0$$

This means that for almost all $\omega \in \Omega$ there exists $k_1(\omega)$ such that for $k \ge k_1$

$$\sup_{0\le t\le T}\left|\left(\int_0^t G_{n_{k+1}} dB - \int_0^t G_{n_k} dB\right)(\omega)\right| \le 2^{-k}, \quad k \ge k_1$$

So a.s. we get uniform convergence. Since $t \to \int_0^t G_n dB$ is continuous, the uniform limit is also continuous. It is a martingale because it is the L_2-limit of martingales. It is clearly square integrable because of the Itô isometry. ■

5.4 Exercises.

1. Given a discrete Martingale $(X_n)_{n\in\mathbb{N}}$, in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{n}})_{\mathrm{n}\in\mathrm{J}}, \mathbb{P})$ and a bounded process $(A_n)_n$ such that A_n is $\mathcal{F}_{n-1}$ measurable, $n = 1, 2, \ldots$ prove that the process
$$(A \cdot X)_n = Z_n = \sum_{i=1}^{n} A_i(X_i - X_{i-1}) = \sum_{i=1}^{n} A_i \Delta X_i$$
is a Martingale and compute its variance.

2. Prove that if $1 \leq p < 2$ the p-variation of the BM is infinite.

3. Prove if a process F is measurable (in the product space $L_2(([0,T] \times \Omega), \mathcal{A}, \mu))$ and adapted then it belongs to $\Gamma_2[0,T]$. Is the recipirocal true?

4. Prove that the a.s limit of a sequence of square integrable martingales (if it exists) is also a martingale.

5. Given a random variable X in $L_1(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$, prove that $X_t := \mathbb{E}(X|\mathcal{F}_t)$ is a Martingale in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$.

6. Prove the formula given in the former section for the Paley -Wiener integral.

7. Prove that the definition of $\int_0^t F dB$ for $F \in \Gamma^2[0,T]$ does not depend on the approximating sequence.

8. Recall de proof of the basic result: if $f_n : [a,b] \to \mathbb{R}$ are continuous functions such that $(f_n)_{n\in\mathbb{N}}$ is a Cauchy sequence for the uniform norm, show that the limit exists and is a continuous function $f : [a,b] \to \mathbb{R}$.

9. Prove that definition (5.8) of the class $\mathcal{E}[0,T]$ can also be stated as one of the following equivalent formulations:

   (a) A process G is in $\mathcal{E}[0,T]$ if there exists a partition $\pi = \{0 = t_0 < t_1 < \cdots < t_m = T\}$ and random variables $G_k : \Omega \longrightarrow \mathbb{R}$ such that each G_k is $\mathcal{F}_k$ measurable, square integrable for $k = 0, 1, \ldots, m-1$ and
   $$G(t,\omega) = G_k(\omega), \quad \text{for} \quad t \in [t_k, t_{k+1}), \quad k = 1, 2, \ldots, m-1.$$

   (b) A process G is in $\mathcal{E}[0,T]$ if it is adapted and there exists a partition $\pi = \{0 = t_0 < t_1 < \cdots < t_m = T\}$ and square integrable random variables $G_k : \Omega \longrightarrow \mathbb{R}$ $k = 0, 1, \ldots, m-1$ such that
   $$G(t,\omega) = G_k(\omega), \quad \text{for} \quad t \in [t_k, t_{k+1}), \quad k = 1, 2, \ldots, m-1.$$

10. Prove that if $G \in \mathcal{E}[0,T]$ and $0 \leq u \leq v \leq T$ then the process
$$G(s,\omega)\mathbb{1}_{[u,v)}(s)$$
is also in $\mathcal{E}^2[0,T]$.

11. Prove that if $G \in \Gamma^2[0,T]$ and $0 \leq u \leq v \leq T$ then the process
$$G(s,\omega)\mathbb{1}_{[u,v)}(s)$$
is also in $\Gamma^2[0,T]$.

12. Prove proposition (5.10) and theorem (5.16).

13. If $F, G \in \Gamma_2[0,T]$ prove that

$$\mathbb{E}\left[\int_0^t GdB \int_0^t FdB\right] = \mathbb{E}\left[\int_0^t GFds\right], \quad t \in [0,T]$$

5.5 Itô Formulas

As we have seen the computation of stochastic integrals is not easy. The Itô formula allows us to compute many more and it is fundamental in the theory, and is the main tool for the study of SDE. We will give the main ideas of two classical forms of proving the theorem.

The first approach to the simplest Itô formula is using the Taylor expansion:

Theorem 5.18. *Given $f\colon \mathbb{R} \to \mathbb{R}$ of class $\mathcal{C}^2$, a real Brownian motion B, for any $t \in (0,T]$,*

$$f(x+B_t) = f(x) + \int_0^t f'(x+B_s)dB_s + \frac{1}{2}\int_0^t f''(x+B_s)ds$$

Proof. We will do the case when f, f' and f'' are bounded by a constant $M \in \mathbb{R}^+$ and are uniformly continuous.
Let π_n be a partition of $[0,t]$ given by $\{0 = t_0^n \leq t_1^n \leq \ldots \leq t_{m_n}^n = t\}$. We can use the Taylor formula to obtain

$$f(x+B_t) = f(x) + \sum_{\pi_n} f'(x+B_{t_j^n})[B_{t_{j+1}^n} - B_{t_j^n}] + \frac{1}{2}\sum_{\pi_n} f''(x+B_{t_j^n})[B_{t_{j+1}^n} - B_{t_j^n}]^2 + R_n.$$

We have to analyze the error term R_n, given by:

$$R_n = \sum_{\pi_n} \left(f''(x+\xi_j^n) - f''(x+B_{t_j^n})\right)[B_{t_{j+1}^n} - B_{t_j^n}]^2$$

where ξ_n^j is a (random) point in the (non ordered) interval $(B_{t_j^n}(\omega), B_{t_{j+1}^n}(\omega))$.

We also use the notations:

$$\begin{aligned}
A_n &= \sum_{\pi_n} f''(x+B_{t_j^n})[B_{t_{j+1}^n} - B_{t_j^n}]^2,\\
B_n &= \sum_{\pi_n} f''(x+B_{t_j^n})[t_{j+1}^n - t_j^n],\\
L_n(j) &= f''(x+B_{t_j^n})[(B_{t_{j+1}^n} - B_{t_j^n})^2 - (t_{j+1}^n - t_j^n)],\\
\Delta B(t_j^n) &= (B_{t_{j+1}^n} - B_{t_j^n}),\ j = 1,2,\ldots,(m_n-1),\\
\Delta t_j^n &= t_{j+1}^n - t_j^n.
\end{aligned}$$

We will prove that

1. $R_n \to 0$ in $L_1(\Omega)$.

2. $\mathbb{E}\left[(A_n - B_n)^2\right] \to 0$.

3. $B_n \to \int_0^t f''(x+B_s)ds$. a.s.

For the first one:

$$\begin{aligned}|R_n| &\leq \sum_{\pi_n} |f''(x+\xi_j^n) - f''(x+B_{t_j^n})| \cdot [\Delta B(t_j^n)]^2 \\ &\leq \sup_i |f''(x+\xi_j^n) - f''(x+B_{t_j^n})| \cdot \sum_{\pi_n} [\Delta B(t_j^n]^2\end{aligned}$$

Using the Cauchy -Schwarz inequality

$$[\mathbb{E}(R_n)]^2 \leq \mathbb{E}\left[\left(\sup_j |f''(x+\xi_j^n) - f''(x+B_{t_j^n})|\right)^2\right] \mathbb{E}\left[\left(\sum_{\pi_n} [\Delta B(t_j^n)]^2\right)^2\right]$$

The uniform continuity of $s \to f''(x+B_s)$, the DCT and the L_2-convergence of the quadratic variation allows us to conclude that R_n converges to 0 in $L_1(\Omega)$, and also in probability. Then by the generalized DCT we get convergence in $L_2(\omega)$.

3.-is immediate from the definition of the usual Riemann integral. Then the convergence in L_2 is a consequence of the DCT.

We now prove 2. First we observe that for $i \neq j$, by independence of the increments:

$$\mathbb{E}\left[L_n(j)L_n(k)\right] = \mathbb{E}\left[\left((\Delta B_{t_j^n})^2 - \Delta t_j^n\right) \cdot \left(\Delta (B_{t_i^n})^2 - \Delta t_i^n\right)\right] = 0$$

$$\mathbb{E}\left[(A_n - B_n)^2\right] = \mathbb{E}\left[\left[\sum_{\pi_n} L_n(j)\right]^2\right] = \mathbb{E}\left[\sum_{j,k} L_n(j)L_n(k)\right]$$

using the former remark, only the terms $j = k$ remain:

$$\begin{aligned}\mathbb{E}\left[\sum_j (L_n(j))^2\right] &\leq M^2 \left[\sum_{\pi_n} \mathbb{E}\left((\Delta B_{t_j^n})^2 - \Delta t_j^n\right)^2\right] \\ &= 2M^2 \sum_{\pi_n} (t_{j+1}^n - t_j^n)^2 \\ &\leq 2M^2 T |\Pi_n|\end{aligned}$$

And this last quantity goes to 0 as the mesh of the partition tends to 0: (We used again the fact that the if $Y \approx \mathcal{N}(0, t_{j+1}^n - t_j^n)$, then $\mathbb{E}(Y^2) = 2(t_{j+1}^n - t_j^n)^2$). ■

The same method can be used to prove a similar result for time dependent functions and of several spatial variables:

Theorem 5.19. *For $f\colon [0,T] \times \mathbb{R}^d \to \mathbb{R}$ of class $\mathcal{C}_2$, $\left\{B_t = (B_t^{(1)}, B_t^{(2)}, \dots B_t^{(d)}) : t \geq 0\right\}$ a d-dimensional Brownian motion and $x \in \mathbb{R}^d$:*

$$\begin{aligned}f(t, x+B_t) = f(0,x) &+ \sum_{j=1}^d \int_0^t f_x'(s, x+B_s) dB_s^{(j)} + \frac{1}{2} \sum_{j=1}^d \int_0^t f_{xx}''(s, x+B_s) ds + \\ &+ \int_0^t f_t'(s, x+B_s) ds.\end{aligned}$$

Proof. We will do it in the case $d = 2$. With the same notations as in the first proof of Itô formula, let Π_n be any partition of $[0,t], t \in [0,T]$. By the method of truncation we can suppose that F and all its derivatives are bounded by $M > 0$.

$$\begin{aligned} f(t, x + B_t) = f(0,x) &+ \sum_{\pi_n} f'_{x_1}(t_j, x + B_{t_j^n})\Delta B^{(1)}_{t_j^n} + \sum_{\pi_n} f'_{x_2}(t_j, x + B_{t_j^n})\Delta B^{(2)}_{t_j^n} + \\ &+ \frac{1}{2}\sum_{\pi_n} f''_{x_1x_1}(t_j, x + B_{t_j^n})[\Delta B^{(1)}_{t_j^n}]^2 + R_n^{(1)} + \\ &+ \frac{1}{2}\sum_{\pi_n} f''_{x_2x_2}(t_j, x + B_{t_j^n})[\Delta B^{(2)}_{t_j^n}]^2 + R_n^{(2)} + \\ &+ \sum_{\pi_n} f'_t(t_j, x + B_{t_j^n})\Delta t_j^n + H_n \end{aligned}$$

The last sum is a Riemann sum for $\displaystyle\int_0^t f'_{x_1}(s, x + B_s)ds$. The other sums and $R_n^{(1)}, R_n^{(2)}$ can be treated as in the first proof of Itô Formula and they give rise to the announced formula. So it remains to prove that the factor H_n goes to 0 as $\lim_n |\Pi_n| = 0$. According to Taylor theorem

$$\begin{aligned} H_n = \frac{1}{2}\sum_{\Pi_n} f''_{x_1,x_2}(a_j)[\Delta B^{(1)}_{t_j^n}]\cdot[\Delta B^{(2)}_{t_j^n}] + \sum_{\Pi_n} f''_{t,t}(a_j)(\Delta t_j^n)^2 + \frac{1}{2}\sum_{\Pi_n} f''_{t,x_1}(a_j)\Delta t_j^n \Delta B^{(1)}_{t_j^n} + \\ + \frac{1}{2}\sum_{\Pi_n} f''_{t,x_2}(a_j)\Delta t_j^n \Delta B^{(2)}_{t_j^n}, \end{aligned}$$

where for each $j = 0, 1, 2, \ldots, (m_n - 1)$ the points a_j, belong to the segment joining $(t_j, x + B_{t_j^n})$ and $(t_{j+1}, x + B_{t_{j+1}^n})$ in the space $\mathbb{R}^2$.

Using the fact that if $X \approx \mathcal{N}(0, \sigma^2)$, then $\mathbb{E}(|X|) = \left(\sqrt{\frac{2}{\pi}}\right)\sigma$ we can prove that each of these terms has an L_2-limit equals to 0 as the norm of the partition goes to 0. For instance

$$\begin{aligned} \mathbb{E}\left[\left(\sum_{\Pi_n} f''_{x_1,x_2}(a_j)[\Delta B^{(1)}_{t_j^n}]\cdot[\Delta B^{(2)}_{t_j^n}]\right)^2\right] &\leq M^2 \sum_{i,j} \mathbb{E}\left[\left|\Delta B^{(1)}_{t_j^n}\Delta B^{(2)}_{t_j^n}\Delta B^{(1)}_{t_i^n}\Delta B^{(2)}_{t_i^n}\right|\right] \\ &= M^2 \sum_{i,j} \frac{2}{\pi}\Delta t_i \cdot \Delta t_j \\ &\leq \left(\frac{2}{\pi}\right)^2 \cdot M^2 |\Pi_n| \cdot T \end{aligned}$$

We used the independence of the increments and of each component to B to obtain that the expectation of product the four increments is equal to the product of the expectations. Then we use the value of $\mathbb{E}\left(\left|\Delta B^{(l)}_{t_k^n}\right|\right)$ for $l = 1, 2,\ \ k = i, j$ and $i, j = 0, 1, 2, \ldots, (m_n - 1)$. ■

Examples 5.20. ♢ *for* $f(x) = x^m,\ \ m \geq 2$ *then* $(B_t)^m = \displaystyle\int_0^t mB_s^{m-1}dB_s + \frac{1}{2}m(m-1)\int_0^t B_s^{m-2}ds$. *So this tells us that* $\displaystyle\int_0^t B_s^{m-1}dB_s = \frac{1}{m}(B_t)^m - \frac{1}{2}(m-1)\int_0^t B_s^{m-2}ds$.

♢ *for* $g(t,x) = tx$, $tB_t = \displaystyle\int_0^t sdB_s + \int_0^t B_s ds$. *This can be done, using the Itô formula in the time dependent case, or it can be proved directly.*

$\diamondsuit$ *for* $u(t,x) = \exp(\lambda x - \frac{\lambda^2 t}{2})$ *and defining the process* $Y_t = u(t, B_t),\ \ t \geq 0$ *we get*

$$\begin{cases} dY = \lambda Y dB \\ Y_0 = 1 \end{cases} \tag{5.2}$$

which is a stochastic differential equation.

5.6 Itô Process

Definition 5.21 (Itô Processes)**.** *Given* $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$*, an* **Itô process** *is a process of the form*

$$X_t = X_0 + \int_0^t A(\omega, s)ds + \int_0^t G(\omega, s)dB_s$$

with

1. $G \in \Gamma_2[0,T]$,
2. A *is a progressive integrable process,*
3. $X_0 \in L_2(\Omega)$ *and* $\mathcal{F}_0$*-measurable.*

In differential form this is written as:

$$dX_t = Ads + GdB$$

More generally:

Definition 5.22 (Itô Processes)**.** *Given* $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$*, an Itô process is a process of the form*

$$X_t = X_0 + \int_0^t A(\omega, s)ds + \sum_{j=1}^m \int_0^t G_j(\omega, s)dB_s^j,\ \ t \in [0,T]$$

with

1. $G_j \in \Gamma_2[0,T],\ \ j = 1, 2, \ldots, m,$
2. A *is a progressive integrable process,*
3. $X_0 \in L_2(\Omega)$ *and* $\mathcal{F}_0$*-measurable,*
4. $B_t = (B_t^{(1)}, B_t^{(2)}, \ldots B_t^{(m)})$ *an* $\mathbb{R}^m$ *-valued Brownian motion.*

In differential form this is written as:

$$dX_t = Adt + \sum_j G_j dB_t^{(j)}.$$

Remark 5.23. *Here we need an admissible filtration.*

This family of processes is a subspace of the set of continuous **semimartingales**, it is the sum of continuous martingales $\int_0^t G_j(\omega, s)dB_s^j$ and a bounded variation continuous process $\int_0^t A(\omega, s)ds$.

There exists an Itô formula for this family of processes, i.e. if we can compute $f(X_t)$ for functions f (with enough regularity conditions). Notice that $(f(X_t) : t \in [0,T])$ is a new process, and it will also be an Itô process as will be seen with the next Itô formula.

Theorem 5.24. *Suppose that X is an Itò process of the form*

$$dX = A\,dt + G\,dB$$

and that $U\colon \mathbb{R}\times[0,T] \to \mathbb{R}$ is of class $\mathcal{C}_2$, then $Y(t) := U(X(t),t)$ has stochastic differential given by

$$\begin{aligned} dU(X,t) &= U_t\,dt + U_x\,dX + \frac{1}{2}U_{xx}G^2\,dt \\ &= (U_t + AU_x + \frac{1}{2}U_{xx}G^2)\,dt + U_xG\,dB \end{aligned}$$

5.7 Integration by parts formula

The former theorem can be proved using theorem **??** or can be proved using the Integration by parts formula:

Theorem 5.25. *Suppose that X_1, X_2 have a stochastic differential*

$$dX_1 = A_1\,dt + G_1\,dB$$

$$dX_2 = A_2\,dt + G_2\,dB$$

with $G_1, G_2 \in \Gamma_2$ and A_1, A_2 progressive and integrable. Then for $s < t$,

$$\int_s^t X_2\,dX_1 = X_1(r)X_2(r) - X_1(s)X_2(s) - \int_s^t X_1\,dX_2 - \int_s^r G_1G_2\,dt$$

This formula is proved first for $\mathcal{F}_0$-measurable r.v, then for elementary processes and finally for processes in $\Gamma_2[0,T]$.

5.8 Exercises

1. Given a discrete Martingale $(X_n)_{n\in\mathbb{N}}$, in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{n}})_{\mathrm{n}\in\mathrm{J}}, \mathbb{P})$ and a bounded process $(A_n)_n$ such that A_n is $\mathcal{F}_{n-1}$ measurable, $n = 1, 2, \ldots$ prove that the process
$$(A \cdot X)_n = Z_n = \sum_{i=1}^{n} A_i(X_i - X_{i-1}) = \sum_{i=1}^{n} A_i \Delta X_i$$
is a Martingale and compute its variance.

2. Prove in detail the result:
   (a) If $1 \leq p < 2$ the p-variation of the BM is a.s. finite.
   (b) The total variation of the paths of BM is a.s. infinite.

3. Given a random variable X in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$, prove that $X_t := \mathbb{E}(X|\mathcal{F}_t)$ is a Martingale in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$. Can you say something about the convergence of (X_t) as $t \longrightarrow \infty$?

4. Prove that definition (**??**) of the class $\mathcal{E}[0, T]$ is equivalent to:
   (a) A process G defined in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$ is in $\mathcal{E}[0, T]$ if there exists a partition $\pi = \{0 = t_0 < t_1 < \cdots < t_m = T\}$ and random variables $G_k : \Omega \longrightarrow \mathbb{R}$ such that each G_k is $\mathcal{F}_k$ measurable and square integrable for $k = 0, 1, \ldots, m-1$ and
$$G(t, \omega) = G_k(\omega), \quad \text{for} \quad t \in [t_k, t_{k+1}), \quad k = 1, 2, \ldots, m-1.$$
   (b) A process G defined in $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t}\in\mathrm{J}}, \mathbb{P})$ is in $\mathcal{E}[0, T]$ if it is adapted and there exists a partition $\pi = \{0 = t_0 < t_1 < \cdots < t_m = T\}$ and square integrable random variables $G_k : \Omega \longrightarrow \mathbb{R}$ $k = 0, 1, \ldots, m-1$ such that
$$G(t, \omega) = G_k(\omega), \quad \text{for} \quad t \in [t_k, t_{k+1}), \quad k = 1, 2, \ldots, m-1.$$

5. Prove proposition (5.10) and theorem (5.16) (except for the last part).

6. If $F, G \in \Gamma_2[0, T]$ prove that
$$\mathbb{E}\left[\int_0^t G dB \int_0^t F dB\right] = \mathbb{E}\left[\int_0^t GF ds\right], \quad t \in [0, T]$$

7. Show, using Itô's formula that
$$Y(t) := e^{B(t) - \frac{t}{2}}$$
solves the stochastic differential equation
$$\begin{cases} dY &= Y dB \\ Y(0) &= 1 \end{cases}$$
(Hint: If $X(t) = B(t) - \frac{t}{2}$ then $dX = -\frac{dt}{2} + dB$).

8. Show that
$$P(t) = p_0 e^{\sigma B(t) + (\mu - \frac{\sigma^2}{2})t}$$
solves
$$\begin{cases} dP &= \mu P dt + \sigma P dB \\ P(0) &= p_0 \end{cases}$$

9. Complete the details of the proof of theorem (5.19)

Chapter 6

Stochastic Differential Equations

We like to find solutions to stochastic differential equations of the form:

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_s.$$

Some examples are the linear equations:

♢ $dX_t = -aX_t dt + \sigma dB_t$. The solution is the Ornstein-Uhlenbeck process

$$X_t = X_0 e^{-at} + \sigma \int_0^t e^{-a(t-s)} dB_s.$$

Since it is the stochastic integral of a deterministic function, we see that is is a Gaussian process: the Riemann approximations of the stochastic integrals Gaussian, so is the limit. We can easily compute the mean function and the covariance matrix of this Gaussian process.

♢ The Vasicek model: $dX_t = (-aX_t + b)dt + \sigma dB_t$ has a solution given by

$$X_t = X_0 e^{-at} + \frac{b}{a}(1 - e^{-at})\sigma \int_0^t e^{-a(t-s)} dB_s.$$

♢ $dZ_t = Z_t \phi(t) dB_t$ with ϕ bounded and progressive. The solution is

$$Z_t = Z_0 \exp\left[\int_0^t \phi(s) dB_s - \frac{1}{2}\int \phi(s)^2 ds\right].$$

♢ $dZ_t = Z_t\left[\mu(t)dt + \phi(t)dB_t\right]$ where μ and ϕ are bounded and progressive and $Z_0 = 1$. The solution is

$$Z_t = \exp\left[\int_0^t \mu(s)ds + \int_0^t \phi(s)dB_s - \frac{1}{2}\int \phi(s)^2 ds\right].$$

What do we mean by a solution? In the space $(\Omega, \mathcal{F}, \mathbb{P})$ we have a Brownian motion B with the canonical filtration. We suppose that b and σ are progressive, such that $b \in L_1(\Omega, \mathcal{F}, \mathbb{P})$ and $\sigma \in \Gamma_2[0, T]$.

Definition 6.1. *A* **strong solution** *of the SDE*

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_s, \;\; X_0 = X(0) \;\; t \in [0, T]$$

with an $\mathcal{F}_0$-measurable initial condition $X(0)$, is an adapted process X w.r.t. the same filtration, verifying the equation

$$X_t = X(0) + \int_0^t b(t, X_t)dt + \int_0^t \sigma(t, X_t)dB_s.$$

Definition 6.2 (Enlarged filtration)**.** *A filtration $(\mathfrak{G}_t : t \in [0, \infty))$ for the d-dimensional Brownian motion $B = (B^1, B^2, \ldots, B^d)$ is called* **admissible** *if*

1. $\sigma(B_s : 0 \leq s \leq t) = \mathcal{F}_t \subset \mathfrak{G}_t$.
2. *for any $0 \leq s < t$, $B_t - B_s$ is independent of $\mathfrak{G}_s$.*
3. *$\mathfrak{G}_0$ is complete i.e. contains all $\mathbb{P}-$null sets.*

Examples 6.3.

1. *The* **completed** *canonical filtration of the Brownian motion.*
2. *$\mathfrak{G}_t := \sigma(B_s : 0 \leq s \leq t, X_0)$ where X_0 is a r.v. independent of the process B, i.e. independent of $\sigma(B_s : 0 \leq s)$ (X_0 will possibly be an initial condition for a SDE).*
3. *If $(\mathfrak{G}_t : t \in [0, \infty))$ is admissible then $(\mathfrak{G}_t^+ = \cap_{\epsilon > 0} \mathfrak{G}_{t+\epsilon}, t \in [0, \infty)$ is also admissible.*

Definition 6.4. *Let $(\Omega, \mathcal{F}, (\mathcal{F}_{\mathrm{t}})_{\mathrm{t} \in \mathrm{J}}, \mathbb{P})$ be a filtered space with an admissible filtration and* $\mathbf{B}$ *is an m-dimensional Brownian motion.*

An $\mathfrak{M}^{n \times m}$-valued process, $\mathbf{G} = (G_{i,j})_{i=1,\ldots,n, j=1,\ldots,m}$ *belongs to $\Gamma_2^{n \times m}[0, T]$ if each $G_{i,j} \in \Gamma^2[0, T]$ for $\{i = 1, \ldots n, j = 1, \ldots m\}$.*

An $\mathbb{R}^n$ valued process $\mathbf{F} = (F_i)_{i=1,\ldots,n}$, *belongs to $\Gamma_1^n[0, T]$ if each component F_i is progressive and verifies:*

$$\mathbb{E}\left[\int_0^T |F_i(s)|ds\right] < \infty.$$

In this case we can construct

$$\int_0^T \mathbf{G} d\mathbf{B}_t.$$

component wise, i.e. the i-th component of this n-dimensional vector is

$$\sum_j \int_0^t G_{i,j} dB_s^j.$$

We also have multidimensional Itô processes defined by

$$X_t = X_0 + \int_0^t \mathbf{F} ds + \int_0^t \mathbf{G} d\mathbf{B}_s.$$

This is an n-dimensional vector with components

$$X_t^i = X_0^i + \int_0^t F^i ds + \sum_j \int_0^t G_{i,j} dB_s^j,$$

$i = 1, \ldots, n$, and we can also prove an Itô Chain rule in this more general context.

$$\int_0^t H \bullet dB = \int_0^t H dB + \frac{1}{2} \int_0^t H_x dt$$

Proposition 6.5 (Tanaka formula and local time).

We would like to find an Itô formula for the not differentiable function $g(x) = |x|$. Define for each $\epsilon > 0$ the function $g_\epsilon(x)$

$$g_\epsilon(x) = \begin{cases} |x| & x \geq \epsilon \\ \frac{1}{2}\left(\epsilon + \frac{x^2}{\epsilon}\right) & x < \epsilon \end{cases}$$

With this g'_ϵ is continuous but g''_ϵ is only piece-wise cat.

$$g'_\epsilon(x) = \begin{cases} 1 & x \geq \epsilon \\ \frac{x}{\epsilon} & x \in (-\epsilon, \epsilon) \\ -1 & x \leq -\epsilon \end{cases} = \begin{cases} \operatorname{sgn}(x) & |x| \geq \epsilon \\ \frac{x}{\epsilon} & x \in (-\epsilon, \epsilon) \end{cases}$$

$$g''_\epsilon(x) = \begin{cases} 0 & |x| \geq \epsilon \\ \frac{1}{\epsilon} & x \in (-\epsilon, \epsilon) \end{cases}$$

($g''_\epsilon(x)$ is discontinuous in a finite number of points), so we can still use the Itô formula and obtain

$$g_\epsilon(B_t) = g_\epsilon(0) + \int_0^t g'_\epsilon(B_s)dB_s + \frac{1}{2}\int_0^t g''_\epsilon(B_s)ds \tag{6.1}$$

first will analyze the last term

$$\begin{aligned} \frac{1}{2}\int_0^t g''_\epsilon(B_s)ds &= \frac{1}{2}\int_0^t \mathbb{1}_{(-\epsilon,\epsilon)}(B_s)\frac{1}{\epsilon}ds \\ &= \frac{1}{2\epsilon}\lambda\left(\{s \in [0,t] : B_s \in (-\epsilon, \epsilon)\}\right) \end{aligned} \tag{6.2}$$

where λ denotes the Lebesgue measure in $\mathbb{R}^+$.
Now we will see what happens with the stochastic integral when ϵ tends to zero

$$\int_0^t g'_\epsilon(B_s)dB_s = \int_0^t \mathbb{1}_{(-\epsilon,\epsilon)}(B_s)\frac{B_s}{\epsilon}dB_s + \int_0^t \operatorname{sgn}(B_s)\mathbb{1}_{|B_s|\geq\epsilon}(B_s)dB_s \tag{6.3}$$

We study the first term on right side. Using the Itô Isometry and Fubini's theorem we obtain

$$\begin{aligned} \mathbb{E}\left[\left(\int_0^t \mathbb{1}_{(-\epsilon,\epsilon)}(B_s)\frac{B_s}{\epsilon}dB_s\right)^2\right] &= \mathbb{E}\left[\int_0^t \mathbb{1}_{(-\epsilon,\epsilon)}(B_s)\frac{B_s^2}{\epsilon^2}ds\right] \\ &= \int_0^t \mathbb{E}\left[\mathbb{1}_{(-\epsilon,\epsilon)}(B_s)\frac{B_s^2}{\epsilon^2}\right]ds \\ &\leq \int_0^t \mathbb{P}\left[\{B_s \in (-\epsilon,\epsilon)\}\right]ds \\ &\leq \int_0^t \frac{2\epsilon}{\sqrt{2\pi s}}ds = C\epsilon\sqrt{t} \longrightarrow 0 \quad (\text{if } \epsilon \longrightarrow 0) \end{aligned}$$

So we conclude that:

$$\int_0^t \mathbb{1}_{(-\epsilon,\epsilon)}(B_s)\frac{B_s}{\epsilon}dB_s \longrightarrow 0 \quad \text{in } L_2(\Omega) \tag{6.4}$$

We study now the second term of right side in (3), using again Itô Isometry and Fubini's Theorem

$$\begin{aligned}\mathbb{E}\left[\left(\int_0^t \left(\operatorname{sgn}(B_s)\mathbb{1}_{|B_s|\geq\epsilon}(B_s) - \operatorname{sgn}(B_s)\right)dBs\right)^2\right] =&\mathbb{E}\left[\left(\int_0^t \operatorname{sgn}(B_s)\mathbb{1}_{|B_s|\leq\epsilon}(B_s)dBs\right)^2\right]\\ =&\mathbb{E}\left[\int_0^t \left(\operatorname{sgn}(B_s)\mathbb{1}_{|B_s|\leq\epsilon}(B_s)\right)^2 ds\right]\\ =&\int_0^t \mathbb{E}\left[\left(\operatorname{sgn}(B_s)\mathbb{1}_{|B_s|\leq\epsilon}(B_s)\right)^2\right] ds\end{aligned}$$

and this integral tends to zero when ϵ tends to zero, therefore we have

$$\int_0^t \operatorname{sgn}(B_s)\mathbb{1}_{|B_s|\geq\epsilon}(B_s)dB_s \xrightarrow{\epsilon\to 0} \int_0^t \operatorname{sgn}(B_s)dB_s \quad \text{in } L_2(\Omega) \tag{6.5}$$

Therefore, since $g_\epsilon(x) \longrightarrow |x|$ point-wise as $\epsilon \to 0$ from equation (6.1) we have (making $\epsilon \longrightarrow 0$)

$$|B_t| = \int_0^t \operatorname{sgn}(B_s)dB_s + L_t(0) \tag{6.6}$$

where $L_t(0) = \lim_{\epsilon\to 0} \lambda\left(\{s \leq t : |B_s| < \epsilon\}\right)$ is called the local time in zero, this limit exists because in equation (6.1) if ϵ tends to zero all other limits exists.

Proposition 6.6. *$\int sgn(B_s)dB_s$ is a Brownian motion (we will use this fact in next section)*

Proof. We define

$$Z_t := \int_0^t \operatorname{sgn}(B_s)dB_s \quad t \geq 0 \tag{6.7}$$

and

$$U(x,t) = \exp\left(i\theta x + \frac{1}{2}\theta^2 t\right) \tag{6.8}$$

Z_t is a Itô Process and $U \in \mathcal{C}_2$ from Theorem (5.24) we have

$$\begin{aligned}U(Z_t,t) =&U(Z_0,0) + \int_0^t \left[\frac{1}{2}\theta^2 U - \frac{1}{2}\theta^2 U(\operatorname{sgn}(B_s))^2\right] ds + \int_0^t i\theta U \operatorname{sgn}(B_s)dB_s\\ =&U(Z_0,0) + \int_0^t i\theta U \operatorname{sgn}(B_s)dB_s \quad (\operatorname{sgn}^2(x) = 1).\end{aligned}$$

then $U(Z_t,t)$ is a martingale. If $0 \leq s \leq t$

$$\begin{aligned}\mathbb{E}\left[U(Z_t,t)|\mathcal{F}_s\right] = U(Z_s,s) \implies& \mathbb{E}\left[e^{i\theta Z_t + \frac{1}{2}\theta^2 t}|\mathcal{F}_s\right] = e^{i\theta Z_s + \frac{1}{2}\theta^2 s}\\ \implies& e^{-i\theta Z_s}\mathbb{E}\left[e^{i\theta Z_t + \frac{1}{2}\theta^2 t}|\mathcal{F}_s\right] = e^{\frac{1}{2}\theta^2 s}\\ \implies& e^{\frac{1}{2}\theta^2 t}\mathbb{E}\left[e^{i\theta(Z_t - Z_s)}|\mathcal{F}_s\right] = e^{\frac{1}{2}\theta^2 s}\\ \implies& \mathbb{E}\left[e^{i\theta(Z_t - Z_s)}|\mathcal{F}_s\right] = e^{-\frac{1}{2}\theta^2(t-s)}\end{aligned}$$

finally, taking expectation in both sides

$$\mathbb{E}\left[e^{i\theta(Z_t - Z_s)}\right] = e^{-\frac{1}{2}\theta^2(t-s)}$$

The left term is the characteristic function of the variable $Z_t - Z_s$, we conclude that $(Z_t - Z_s) \sim N(0, t-s)$, also we have $Z_0 = 0$ and for $0 \leq s \leq t$ $(Z_t - Z_s) \perp Z_s$ because

$$Z_t - Z_s = \int_s^t \operatorname{sgn}(B_{\tilde{s}}) dB_{\tilde{s}} \perp \int_0^s \operatorname{sgn}(B_{\tilde{s}}) dB_{\tilde{s}}$$

Then from definition (4.6) we have that Z_t is a Brownian motion. ■

6.1 Weak Solutions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space and B a Brownian motion with the canonical filtration, suppose that we have $b(x,t)$, $\sigma(x,t) \colon [0,T] \times \mathbb{R} \longrightarrow \mathbb{R}$ and the Stochastic Differential Equation

$$(^*) \begin{cases} dX_t &= b(X_t, t)dt + \sigma(X_t, t)dB_t \\ X_0 &= x_0 \end{cases}$$

We look for another probability space $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathbb{P}})$, processes $(\hat{X}_t, \hat{B}_t)_t$, and a filtration $(\mathcal{H}_t)_t$ where all is well define, and such that $\hat{X}_t$ is $\mathcal{H}_t$-progressive and adapted, $(\hat{B}_t)$ is a $\mathcal{H}_t$-Brownian motion and $\mathcal{H}_t$ is complete, cad. and contains to $\hat{X}_0$. Then we say that $(\hat{X}_t, \hat{B}_t)_t$ is a weak solution of $(*)$ if $\hat{X}_t, \hat{B}_t$ satisfies

$$\begin{cases} d\hat{X}_t &= b(\hat{X}_t, t)dt + \sigma(\hat{X}_t, t)d\hat{B}_t \\ \hat{X}_0 &= \hat{x}_0 \end{cases}$$

This concept is very important and useful because there are stochastic differential equations which have no strong solutions but have a weak solution

Examples 6.7. *Consider the stochastic differential equation*

$$(p) \begin{cases} dX_t &= sgn(B_t)dB_t \\ X_0 &= 0 \end{cases}$$

where B_t is a Brownian motion and $\mathcal{F}_t$ the canonical and completed filtration. Let $\hat{B}_t$ another standard Brownian motion and $\hat{\mathcal{F}}_t$ the canonical filtration, we define

$$Y_t = \int_0^t sgn(\hat{B}_s) d\hat{B}_s.$$

By Tanaka formula

$$Y_t = |\hat{B}_t| - \hat{L}_t(0),$$

then, Y_t is measurable with respect to $\mathcal{G}_t = \sigma\left(|\hat{B}_s| \colon s \leq t\right)$, we have

$$\begin{aligned} \mathcal{G}_t &\subsetneq \hat{\mathcal{F}}_t \\ \therefore \quad \sigma(Y) &\subseteq \mathcal{G}_t \subsetneq \hat{\mathcal{F}}_t \end{aligned}$$

by proposition (6.6) X_t is a Brownian motion, then we can rewrite

$$dB_t = sgn(X_t)dX_t$$

then

$$\hat{\mathcal{F}}_t \subsetneq \sigma(X_s) = \sigma(B_s)$$

but this contradicts that X_t is a strong solution, therefore strong solution do not exist. A weak solution is any standard Brownian motion.

Proposition 6.8. *Any two weak solutions are identical in law, i.e., have the same finite-dimensional distributions.*

Proof. ■

6.2 Exercises

1. Use that
$$Y := e^{\int_0^t g dB - \frac{1}{2}\int_0^t g^2 ds}$$
satisfies
$$dY = gY dW$$
to prove
$$\mathbb{E}\left[e^{\int_0^T g dB}\right] = e^{\frac{1}{2}\int_0^T g^2 ds}$$

2. Let $u = u(x,t)$ be a smooth solution of the *backwards diffusion equation*
$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial^2 u}{\partial x^2} = 0,$$
and suppose $B(\,\cdot\,)$ is a one-dimensional Brownian motion. Show that for each time $t > 0$:
$$\mathbb{E}\left[u\left(B(t), t\right)\right] = u(0,0).$$

3. Calculate $\mathbb{E}\left[B^2(t)\right]$ for the Brownian bridge $B(\,\cdot\,)$, and show in particular that $\mathbb{E}\left[B^2(t)\right] \longrightarrow 0$ as $t \longrightarrow 1^-$.

4. Let X solve the Langevin equation, and suppose that X_0 is an $N\left(0, \frac{\sigma^2}{2b}\right)$ random variable. Show that
$$\mathbb{E}\left[X(s)X(t)\right] = \frac{\sigma^2}{2b}e^{-b|t-s|}$$

5. (a) Consider the ODE
$$\begin{cases} \dot{x} &= x^2 \quad (t > 0) \\ x(0) &= x_0 \end{cases}$$
Show that if $x_0 > 0$, then the solution *blows up to infinity* in finite time.

   (b) Next, look at the ODE
$$\begin{cases} \dot{x} &= x^{\frac{1}{2}} \quad (t > 0) \\ x(0) &= 0 \end{cases}$$
Show that this problem has infinitely many solutions. (Hint: $x \equiv 0$ is a solution. Find also a solution which is positive for times $t > 0$, and then combine these solutions to find ones which are zero for some times and then become positive.)

6. (a) Use the substitution $X = u\left(B\right)$ to solve the SDE
$$\begin{cases} dX &= -\frac{1}{2}e^{-2X}dt + e^{-X}dB \\ X(0) &= x_0 \end{cases}$$

   (b) Show that the solution blows up, at a finite random time.

7. Solve the SDE $dX = -Xdt + e^t dB$.

8. Solve

$$\begin{cases} dX &= \frac{1}{2}\sigma'(X)\,\sigma(X)\,dt + \sigma(X)\,dB \\ X(0) &= 0 \end{cases}$$

where B is a one-dimensional Brownian motion and σ is a smooth, positive function. (Hint: Let $f(x) := \int_0^x \frac{dy}{\sigma(y)}$ and set $g := f^{-1}$, the inverse function of f. Show $X := g(B)$.)

9. Compute $\mathbb{E}[|B_t|]$

10. If (B_t) is a one-dimensional Brownian motion, say whether or not the following processes are martingales or semimartingales:

$$(B_t + 4t)_t\,, \quad \left(B_t^2\right)_t\,, \quad \left(t^2 B_t - \int_0^t sB_s ds\right)_t\,, \quad \left(B_t^3 - 3tB_t - t\right)_t\,, \quad (tB_t - B_t)_t\,.$$

11. If $\mathbf{B}_t = \left(B_t^{(1)}, B_t^{(2)}\right)_t$ is a two-dimensional Brownian motion, is $Z_t = B_t^{(1)} \cdot B_t^{(2)}$ a martingale?

12. Suppose that $F \in \Gamma_2[0,T]$ has the following property

$$\mathbb{E}\left[|F(s,\cdot) - F(t,\cdot)|^2\right] \leq K|s-t|^{1+\epsilon}\,, \quad s,t \in [0,T]$$

for some $K, \epsilon > 0$. Then given a partition $\Pi_n : 0 = t_0 < t_1 < \cdots < t_m = t$ for the interval $[0,t]$, $0 < t < T$ and taking **any point** $s_i \in [t_i, t_{i+1})$ the Riemann sums

$$S_n := \sum_j F(s_i,\cdot)\Delta B_{t_j}$$

converge in the L_2-norm to $\int_0^t F dB$. Why can you take any intermediate point in this case? Compare with lemma (**??**). Is there a contradiction?

13. Is the Orstein-Uhlenbeck process

$$X_t = X_0 e^{-at} + \sigma \int_0^t e^{-a(t-s)} dB_s$$

a martingale?, a semimartingale?

Prove that it is a Gaussian process, compute its mean function $m_t = \mathbb{E}[X_t]$ and its covariance function $C(t,s)$ $t,s \in [0,T]$

14. Give conditions on $f : \mathbb{R}^+ \times \mathbb{R}^d \longrightarrow \mathbb{R}$ so that process $[f(t,B_t)]_t$ be a martingale.

Bibliography

[1] Lawrence C. Evans : An introduction to Stochastic differential equations. *AMS* (2013).

[2] Bernt Oksendal: Stochastic differential equations.*Springer* (2000)

[3] D. Revuz, M. Yor : Continuous Martingales and Brownian motion. *Springer* (1999)

[4] René L. Schilling, Lothar Partzsch: Brownian motion. *De Gruyter* (2012.)